

Computational Analysis of Metal-Dependent Structural
Variations of Acireductone Dioxygenase Enzymes

by
Andrew Fisher

A thesis submitted in partial fulfillment of the requirements
for graduation in Chemistry.

Whitman College

2026

Certificate of Approval

This is to certify that the accompanying thesis by Andrew Fisher has been accepted in partial fulfillment of the requirements for graduation with a degree in Chemistry.

Dalia R. Biswas

Whitman College

May 17, 2026

Table of Contents

Table of Contents.....	iii
Acknowledgements.....	iv
Abstract.....	v
List of Tables.....	vi
List of Figures.....	vii
Chapter 1 - The Acireductone Dioxygenase Family of Enzymes	
Introduction.....	1
Computational Methods.....	10
Computational Workflow.....	10
Results & Analysis.....	13
Conclusions & Future Directions.....	20
Appendices.....	22
A: Annotated ORCA Slurm File.....	22
B: Annotated ORCA Input File for Geometry Optimizations.....	23
C: Individual Structural Models, Modeled in Pymol.....	24
D: Annotated LargePrint Input File for SPE Calculations.....	32
E: Annotated LargePrint Parser for SPE files.....	33
F: Map of Individual Active Site Spin Populations from LargePrint Parser.....	41
G: Map of Individual Mulliken Atomic Charges from LargePrint Parser.....	45
Bibliography.....	49

Acknowledgements

I wish to express my wholehearted gratitude to Dr. Dalia Biswas for the opportunity to join her research team in my senior year and for her constant encouragement and mentorship throughout my time at Whitman. Her advice spanned from streamlining our workflow to digesting the literature within the framework of my undergraduate studies. Our partnership with Dr. Santiago Toledo's lab at American University was pivotal in providing an experimental basis for this research and in the guidance and support from Dr. Toledo himself. I would like to extend my gratitude to Professor Jonathan Collins for his help in student counseling and facilitating my studies in chemistry, as well as to Professor Machonkin for his insights into computational techniques and inorganic bonding.

Abstract

Acireductone dioxygenase (ARD), a key enzyme involved in the methionine salvage pathway (MSP), converts acireductone, a product of the disposal of the polyamine inhibitor methylthioadenosine (MTA), into usable methionine. ARD displays dual functionality depending on the divalent metal ion bound at its active site, catalyzing acireductone along distinct pathways, yielding different products. With Fe-bound, ARD channels acireductone through the canonical MSP, regenerating methionine. In contrast, Ni-substitution redirects this reaction to an alternate oxidative pathway, producing formate, methylthiopropionate, and carbon monoxide. To investigate metal-dependent structural differences, we conducted a comparative analysis of published crystal structures using computational tools and will present our insights.

List of Tables

Table 1. Metals identified in ARD and their published reactivity relative to ARD's dual chemistry in the methionine salvage pathway (MSP)

Table 2. NPA charges throughout the charge hypothesis' proposed reaction, adapted from ref. 12

Table 3. PDB-published ARD structures sampled for optimization

Table 4. Arithmetic means of RMSD values for each non-backbone residue included in the calculations, as well as their sum.

Table 5. Overall structural RMSD values between each optimized structure calculated using PyMol without outlier rejection, as well as the Ensemble RMSD.

Table 6. Calculated single point energies, S2, optimized spin multiplicities and the spin populations of the metal and whole active site expressed as a percentage of total spin

Table 7. Calculated DFT spin populations of metals & ligating atoms

Table 8. Calculated Mulliken atomic charges of metals & ligating atoms

List of Figures

- Fig 1). Comparison of N_2O and N_3O mononuclear ferrous binding centers.
- Fig 2). The Methionine Salvage Pathway featuring both isoforms of Ko-ARD, adapted from ref.1
- Fig 3). Hypothesized reaction of Fe-ARD' featuring a 5-membered ring coordination to the substrate in *Klebsiella oxytoca*'s Fe-ARD, adapted from ref. 12
- Fig 4). Small model simulations of ARD reactivity derived by X-ray crystallography, adapted from ref. 6
- Fig 5). Proposed reactions of Ko-ARD' with 6-membered ring coordination, adapted from ref. 12
- Fig 6). Reaction profile of transition state energies between different Co(II) spin states in Co-2HJI, relative to doublet state [kcal/mol], adapted from ref. 15
- Fig 7). Summarized computational workflow showing the differences between NMR and XRD structures and the cyclic modifications of our models, modelled with PyMol & Chemcraft.
- Fig 8). Predetermination of high spin, octahedral M(II) spin states by M.O. analysis
- Fig 9). Model Energy per Step N of Normal SCF convergences in ORCA illustrating minimal metal-dependence of optimization, but large variance between workflows.
- Fig 10). RMSD formula for cartesian drift of atomic positions, adapted from ref. 25
- Fig 11). Visualization of water molecule displacement in 2HJI and Co-2HJI from their metal cofactors and onto arginine 154.
- Fig 12). Illustration of Arg154's optimized position (L-R) in the presence of a substrate analog, in the presence of Trp155, and in the absence of neither, modeled with Chemcraft.

Fig 13). Comparison of Optimized Geometries of Mm-ARD (Ni, Co and Fe), modeled with PyMol

Fig 14). Optimized geometries of NMR-solved Ko-ARD (Ni, Co and Fe), modeled with PyMol

The Acireductone Dioxygenase Family of Enzymes

Dioxygenase enzymes facilitate a diverse array of high-energy redox reactions on organic substrates by incorporating dioxygen at their active sites, often mediated by transition-metal cofactors (Deshpande *et al.*, 2017). These metal cofactors enable the circumvention of traditional organic spin barriers, thereby allowing for the activation of dioxygen in the presence of a substrate or, conversely, the activation of the substrate by dioxygen. It is this catalytic versatility that enables biological processes ranging from cellular respiration to regulatory mechanisms. While iron has been observed to be most prevalent in oxygen related enzymes, due to its widespread bioavailability, dioxygenases exhibit a robust ability to accommodate various metal ions, with patterns of promiscuity largely reflecting biological metal concentrations. Within the class of iron-containing dioxygenases, further subdivisions exist, and of note for our work is mononuclear iron dioxygenases, those where a single ferrous center acts on a single substrate.

These mononuclear iron dioxygenases are extraordinary even among metal enzymes due to their significant diversity in their substrate specificity, reactivity, and binding motifs across and even within species. Among these motifs, the facial 2-His-1-Carboxylate motif is noticeably prevalent, having been identified in over 100 ferrous metalloenzymes, regardless of sequence homology (Hegg & Que, 1997; Kal & Que, 2017). Investigations into these N₂O (two nitrogen, one oxygen) motifs have demonstrated their exceptional flexibility, allowing substrate binding at any of the three available facial coordination sites. In the absence of substrate, these sites are commonly occupied by labile water molecules *in vivo*, both conferring stability in aqueous environments and enabling easy displacement by incoming reactants. This facial triad also enables direct dioxygen activation by the captured metal ion, as through the substitution of water

by substrate, the fragile network of hydrogen bonds between the labile waters is broken, revealing previously inaccessible redox properties of the cofactor.

Although N₂O coordination is predominant among ferrous metalloenzymes, the same cannot be said for N₃O coordination motifs, wherein the incorporation of an additional histidine residue into the metal coordination sphere might be expected to enhance stability in geometric and cofactor spin, empirical evidence suggests that such complexes exhibit reduced if not completely altered reactivity (Hegg & Que, 1997). These N₃O-coordinated iron centers are generally characterized by reduced flexibility in substrate binding and their capacity for dioxygen activation, as demonstrated by studies showing that FeSOD and SLO-1 are explicitly incapable of directly activating dioxygen (Fadul *et al.*, 2024).

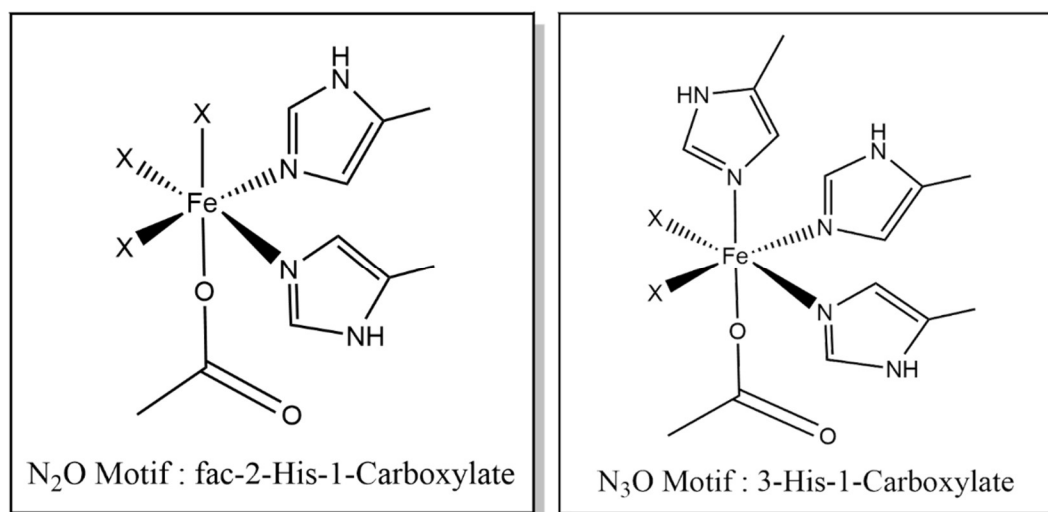


Fig 1. Comparison of N₂O and N₃O mononuclear ferrous binding center

Subsequent research has since elucidated the potential for direct oxygen activation by certain N₃O metal centers, most notably in the highly conserved active site of acireductone dioxygenase (ARD) enzymes from the β -cupin family (Chavez *et al.*, 2011). ARD is present in both mammalian and bacterial methionine salvage pathways (MSP) and catalyzes the oxidative

cleavage of 1,2-dihydroxy-2-keto-5-methylthiopentene (acireductone) via two distinct pathways, the outcome of which is determined exclusively by the identity of the resident divalent metal cofactor (Deshpande *et al.*, 2016; Miłaczewska *et al.*, 2018). This dual reactivity enables ARD to direct acireductone either along the canonical MSP, yielding 2-keto-4-methylthiobutyrate (KMTB) and formate, or along an off-pathway shunt, producing 3-methylthiopropionate (MTP), carbon monoxide, and formate (Fig 2). To date, 12 ARD structures have been deposited in the Protein Data Bank (PDB), incorporating metals such as iron, nickel, cobalt, and cadmium; however, ARD has also been observed to facilitate reactivity by utilizing magnesium and manganese (Deshpande *et al.*, 2017). Since the dual chemistry of ARD was initially characterized in *Klebsiella oxytoca*, this work refers to all on-pathway reactions as the native Fe-ARD activity of *Klebsiella oxytoca* and off-pathway reactions as Ni-ARD activity. The discovery of ARD's dual function has prompted extensive discussion regarding the evolutionary pressures that may have shaped such specialized reactivity, including the synthesis of S-adenosylmethionine (SAM) and polyamines in mammals via Fe-ARD activity.

Table 1. Metals identified in ARD and their published reactivity relative to ARD's dual chemistry in the methionine salvage pathway (MSP)

Metal (2+)	Origin	PDB ID	Reactivity
Mg	Origin	N/A	On
Fe	<i>K. oxytoca</i>	2HJI	On
Co	<i>M. musculus</i>	5I8Y	Off
Ni	<i>K. oxytoca</i> , <i>M. musculus</i>	1ZRR, 5I8S	Off
Mn	N/A	N/A	Off
Cd	<i>B. anthracis</i>	4QGM	N/A

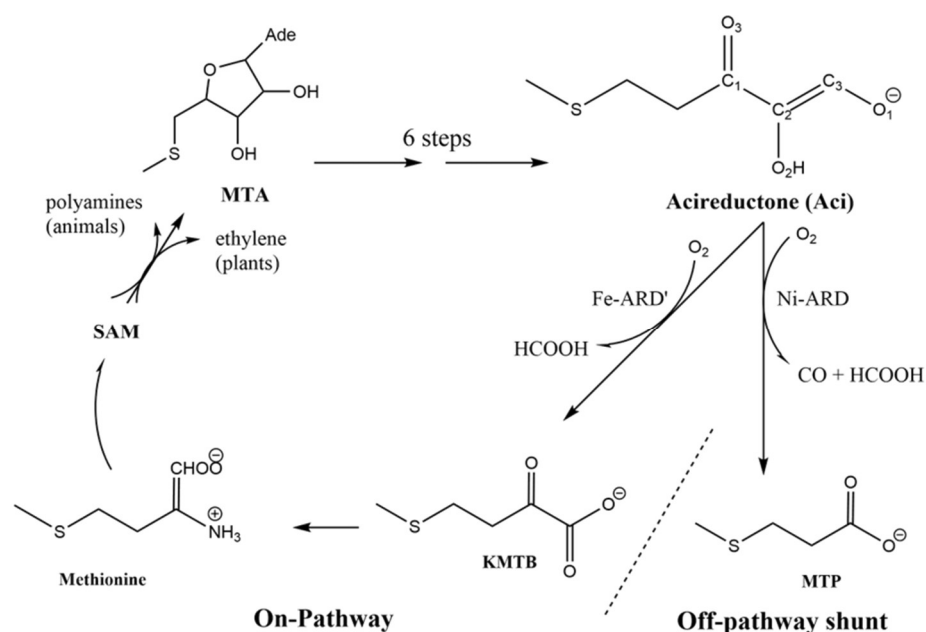


Fig 2. The Methionine Salvage Pathway featuring both isoforms of Ko-ARD, adapted from ref. 1

Of note to evolution, polyamines remain implicated in the dysregulation of cellular proliferation and have therefore been postulated to contribute to carcinogenic or metastatic processes. Elevated polyamine levels, as produced via the on-pathway MSP, can inhibit apoptotic signaling, an effect that has been described as a possible evolutionary rationale for the off-pathway shunt (Fadul *et al.*, 2024), wherein the organism may have some control over the arrest of cellular reproduction. Further, on-pathway Fe-ARD activity has been shown to modulate matrix metalloproteinase 14 (MT1-MMP), a transmembrane protease associated with tumor metastasis. While off-pathway ARD activity has not been unequivocally demonstrated in humans, its presence in murine models suggests that related isozymes could represent viable targets for cancer therapy, though the toxicity of nickel in mammalian systems could require the implementation of another off-pathway metal, as observed in *Mus musculus*' cobalt-native 5I8S (Table 1) (Deshpande *et al.*, 2017).

Though metalloenzymes are well known for their promiscuity in metal binding, reactivity is typically ascribed to structural differences within the protein scaffold (either primary or tertiary). This phenomenon of solely metal-specific regioselectivity in isozymes is particularly unique to ARD and provides a valuable model for investigating the influence of transition-metal identity on enzymatic function. Since its discovery, ARD has attracted considerable attention from both microbiologists and inorganic chemists, resulting in a substantial body of literature exploring the underlying mechanisms of ARD's dual reactivity.

Insights from Geometry and Reactivity

The most cited work concerning ARD's native dual reactivity originates in docking simulations of NMR-resolved bacterial protein structures conducted by Dai *et al.*, which led to the development of the chelate hypothesis for 1ZRR and 2HJI. According to these studies, the observed differences in reactivity between the native iron- and nickel-bound forms of ARD in *Klebsiella oxytoca* are attributed to metal-dependent variations in their covalent E-S complex, stabilized and modulated by structural features and steric constraints within the active site (Fig 3). While the proposal of a five-membered ring coordination in Fe-ARD was later criticized for its perceived instability by sterics, the foundational work of Dai *et al.* prompted further investigation (Dai *et al.*, 1999).

Much of the initial criticism stemmed from small-molecule in vitro simulations of the ARD active site, which provided experimental evidence that both iron and nickel can adopt six-

membered ring coordination geometries upon substrate binding. Uniquely, however, these studies illustrated an incapacity of Fe-ARD to catalyze the on-pathway reaction in anhydrous solvents, wherein it facilitated Ni-ARD chemistry exclusively (Fig 4). Only when water was introduced in vitro did their distinct reactivity patterns emerge, leading to the hypothesis that not only does water influence the reaction outcome, but the metal ions themselves influence water's accessibility within the active site (Allpress *et al.*, 2012).

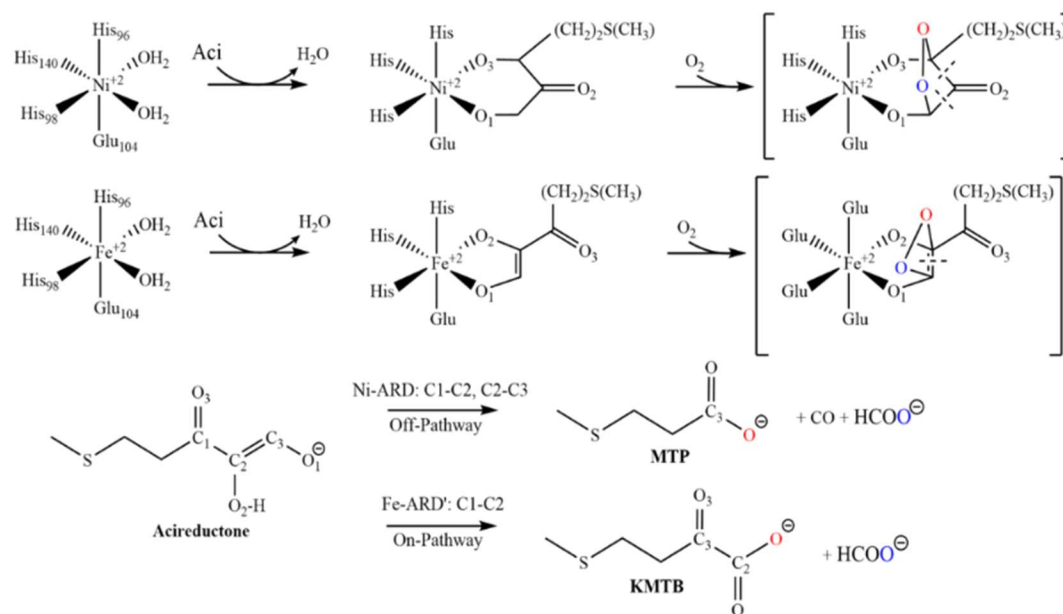


Fig 3. Hypothesized reaction of Fe-ARD' featuring a 5-membered ring coordination to the substrate in *Klebsiella oxytoca*'s Fe-ARD, adapted from ref. 12

Sparta *et al.*'s computationally derived charge-transfer hypothesis employed their lab's novel hybrid quantum mechanics/discrete molecular dynamics (QM/DMD) approach to parse the entire ARD protein and quantify transition-state energies in the active site (Sparta *et al.* 2013). Their findings corroborated the plausibility of binding through a six-membered ring E-S conformation for both Fe and Ni-ARD and highlighted the importance of a conserved arginine residue at the periphery of the active site (Table 2). Using their computational models, Sparta *et*

al. documented that the six-membered ring configuration is energetically favored in both Fe-ARD and Ni-ARD, as well as a potential second transition state specific to Fe-ARD after the cleavage of dioxygen (Fig 5).

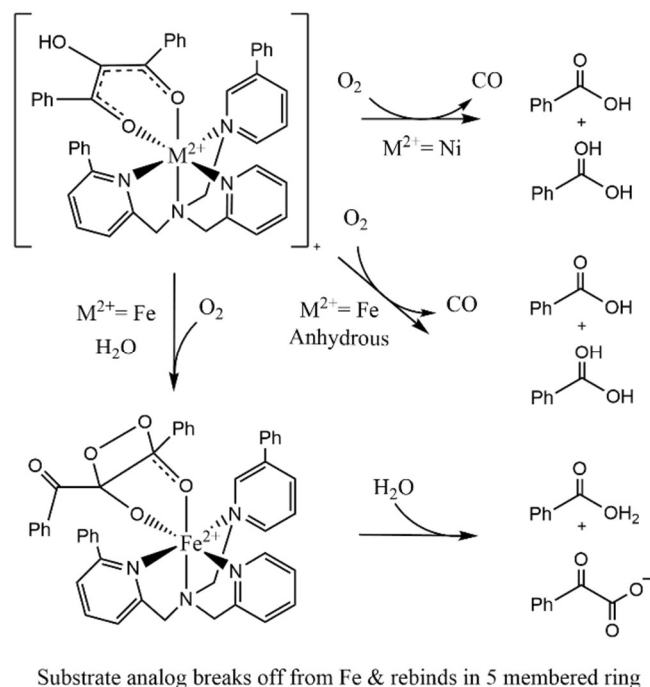


Fig 4. Small model simulations of ARD reactivity derived by X-ray crystallography, adapted from ref. 6

Then, through stepwise natural population analysis (NPA), Sparta *et al.* expanded their intrigue into their model's capacity for M-L charge transfer, revealing that the Fe(II) center facilitates significant transfer of negative charge from the substrate to surrounding protein residues, whereas the Ni(II) center acts predominantly as a strict Lewis acid. These disparities in intrinsic electronic traits, though predominantly reported in catalytic studies, such as those on Ni-Fe mixed-metal oxides, certainly provide evidence supporting Sparta's findings (Bates *et al.*, 2016; Miłaczewska *et al.*, 2018).

Table 2. NPA charges throughout the charge hypothesis' proposed reaction, adapted from ref. 12

Table 5: NPA charges of 1ZRR active site through Ni-ARD reaction.

Step	Ni ²⁺	Aci	O ₂
S Binding	1.40	-0.97	-0.65
‡ ₁	1.40	-0.95	-0.67
Product	1.40	-0.22	-1.45

Table 6: NPA charges of 2HJI active site through current Fe-ARD' reaction.

Step	Fe ²⁺	Aci	O ₂
S Binding	1.46	-1.00	-0.66
‡ ₁	1.45	-0.98	-0.68
O ₂ split	1.48	-0.66	-1.30
‡ ₂	1.50	-0.65	-1.34
Product	1.47	-0.33	-1.41

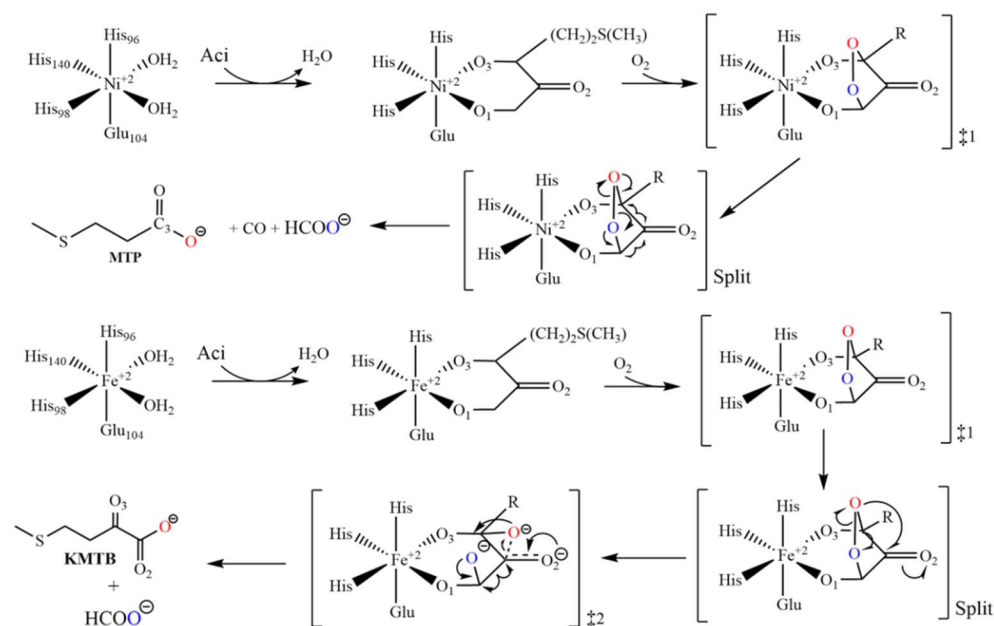
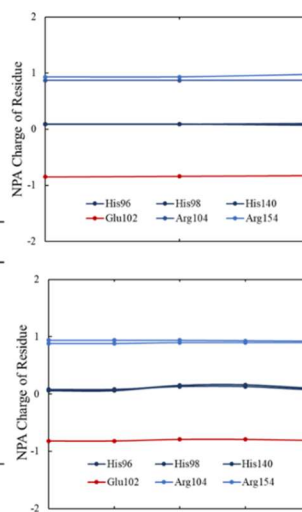


Fig 5. Proposed reactions of Ko-ARD with 6-membered ring coordination, adapted from ref. 12

In a later publication by Sparta *et al.*, they noted that in their model, although the six-membered ring conformation was energetically preferred for both Fe-ARD and Ni-ARD, the observed reactivity was highly sensitive to the metal's spin state. Specifically, they proposed that Fe-ARD could, in theory, emulate Ni-ARD reactivity when constrained to the singlet state,

despite its N3O-bound, native quintet ground state. This work also included silico metal-swapping experiments, demonstrating that Co-ARD, while experimentally producing exclusively off-pathway reactivity, could in principle support on-pathway chemistry in both high- and low-spin states (Fig 6). Nonetheless, the absence of experimental evidence for spin-state interconversion in ARD, coupled with the proposed reactivity of the high spin quartet ground state expected for Co(II) in octahedral environments, highlights the limitations of current computational models, particularly regarding DMD's weaknesses in calculating spin populations (Valdez *et al.*, 2014). Such discrepancies clarify the intense precautions necessary in QM technique selection, as the oversimplified potential energy functions in DMD can lead to staggering inaccuracies in ligand binding energies, trapping simulations in localized, low energy minima and limiting their ability to reach a true energetic minimum.

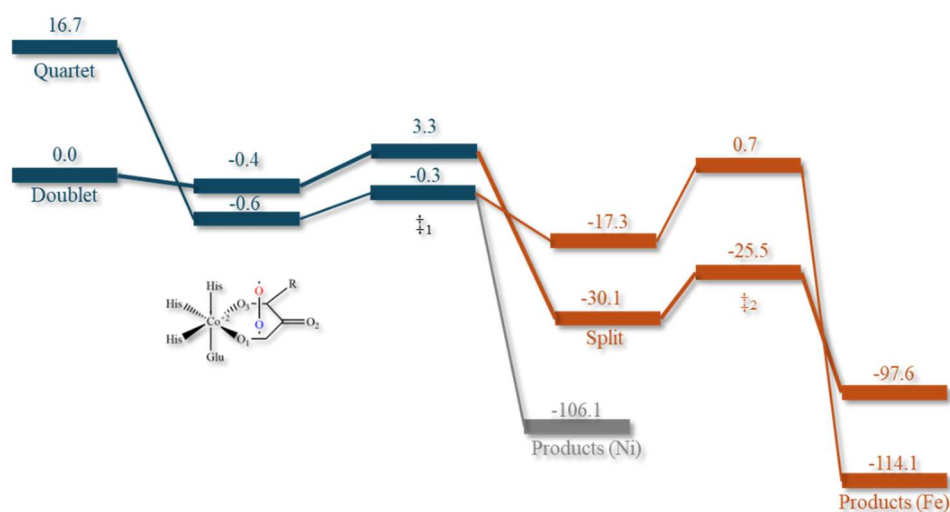


Fig 6. Reaction profile of transition state energies between different Co(II) spin states in Co-2HJI, relative to doublet state [kcal/mol], adapted from ref. 1

Given that two of the main standing hypotheses for ARD reactivity are directly linked to metal effects on structure, we settled on a DFT geometry optimization of an abbreviated model

of four published ARD structures as a compromise between accuracy and computational cost. With these optimized structures in the absence of substrate, we would then have an improved model with which to test for metal dependence in the structure-function relationship in the presence of substrate, as well as better measure the energies of subsequent transition states.

Computational Methods

All ARD structures were downloaded from the Protein Data Bank by PyMol, where the majority of the modeling and RMSD calculations were performed. DFT geometry optimizations were performed with ORCA using bp86 and the def2-tzvp for heavy atoms except C and H. Minimal basis set for light atoms (def2-svp) were used to speed up the calculation. Single point energies were calculated subsequently with b31yp with the same basis set. ChemCraft and PyMol were used for data analysis and visualization.

Computational Workflow

All protein structures were imported from the Protein Data Bank to PyMol and xyz coordinates of all models were extracted for DFT calculations. To ensure that our structures constituted a representative sample of published native ARD structures and their reactivity, only first-row transition metals Fe(II), Co(II), and Ni(II) were included in our manifest (Table 3). Upon loading these PDBs into PyMol, we observe remarkable structural homology between the proteins, but large differences in how their files are stored in the software. Proteins whose structures are solved by protein NMR have a set number of solution states (e.g., 1ZRR has 17 possible states) to choose between, whereas X-Ray structures are completely deprotonated.

Table 3. PDB-published ARD structures sampled for optimization

PDB ID	Origin	Metal (2+)	Reactivity	Structure
1ZRR	K. oxytoca	Ni	Off	NMR
2HJI	K. oxytoca	Fe	On	NMR
5I8S	M. musculus	Ni	Off	X-Ray
5I8Y	M. musculus	Co	Off	X-Ray

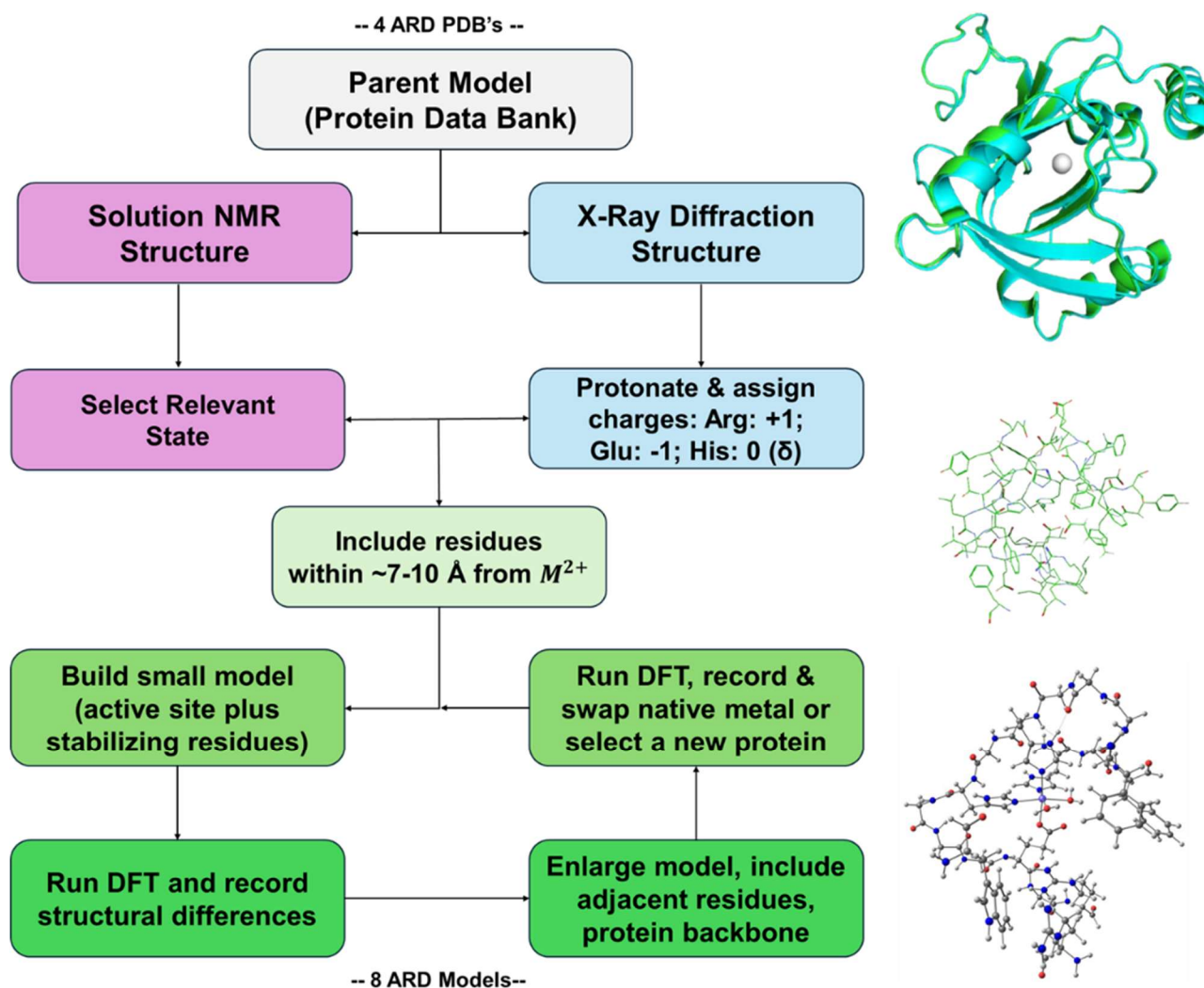


Fig 7. Summarized computational workflow showing the differences between NMR and XRD structures and the cyclic modifications of our models, modelled with PyMol & Chemcraft.

In terms of the divergence between these workflows, it is critical to keep in mind that X-ray structures are merely a snapshot, a weighted average of what the protein should look like, notably not including hydrogen. For this reason, before running our calculations, we must reprotonate each atom and thus intuit it's charge. On the other hand, NMR solved proteins are dynamic, arranging an ensemble of selected states from which to choose. For NMR, we define “relevant” or “reasonable” states as those with a strong active-site presence of the conserved arginine residues, those thought to have some role in the deprotonation of the substrate or in providing charge stability through the transition states (Sparta *et al.*, 2013). We then select all residues within 7-10 angstroms of the metal cofactor, aiming to build a representative yet computationally affordable structure with which to run our calculations. Still, these structures are often too large for our supercluster, so we replace all “irrelevant” side chains (those facing away from the active site) with hydrogens, effectively replacing them with glycine. All alpha-carbon-hydrogen atoms are then frozen via Cartesian constraints, the optimization level is set, and the structure's net charge and M(II) spin states are calculated by hand through analysis of the metal's octahedral antibonding orbitals (Fig 8).

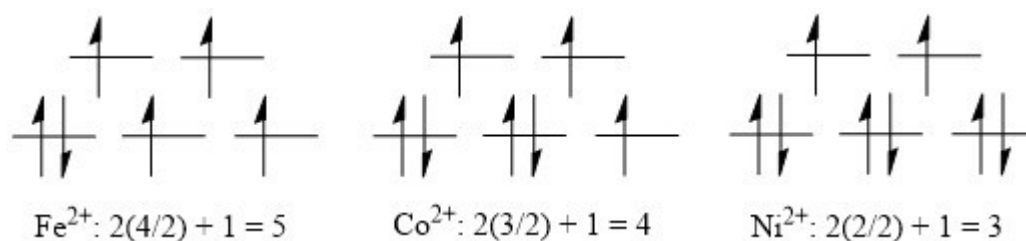


Fig 8. Predetermination of high spin, octahedral M(II) spin states by MO analysis. Spin multiplicity ($2S+1$) is shown in the figure.

Now, we can proceed with chemistry inaccessible via experimental methods by swapping the metal's identity in the input file and recalculating its spin by our M.O. analysis. We then run

DFT calculations on different metals in identical structures. Starting with the two NMR/Ko-ARD structures, one Ni(II) and one Fe(II), we swap these metals into two new unique input files with Co(II) to make Co-1ZRR and Co-2HJI. Doing the same with the remaining two X-Ray Diff./Mm-ARD structures, occupied by Ni(II) and Co(II), we swap them both with Fe(II) to make Fe-5I8S and Fe-5I8Y, respectively. Now, from our four starting PDB structures, we have eight final input files ready to be run by DFT and compared via PyMol (Fig. 7). As discussed in our introduction on behalf of the charge transfer hypothesis, DFT can struggle with trapping in localized energy minima, so we chose to break the optimization up into three steps of progressively more rigorous/expensive optimization, starting first with a normal SCF, followed by tight SCF and lastly with a minimally invasive implicit solvent model, all done with a d3bj method, def2-tzvp and def2-svp basis set. Now with our eight optimized structures, we run a DFT single point energy calculation with Large Print to sample spin populations and Mulliken atomic charges using b31yp.

Results & Analysis

It's critical to examine how our workflow affects the outcomes of our research, as although our structural optimizations resulted in adjacent Normal SCF final energy minima, there was a marked difference in the optimization graphs of NMR vs X-ray structures, which was interestingly independent of metal identity. Upon modeling the optimization curves via Chemcraft, we can see that this was largely due to full revolutions of mobile residues such as the arginine and the rotation of the glutamate ligating the metal in the N3O motif, observed only in NMR solved structures (Fig 8). We attribute this largely to the implicit error in "relevant" state selection, as although the protein can be natively found in these states, NMR structures are not necessarily the weighted average structures that concede reactivity.

Root mean square deviation is a mathematical technique for comparing Cartesian coordinates in space, directly calculable via PyMol's integrated RMSD features or via direct comparison of residue coordinates during the optimization steps (Fig 9; Kuzmanic *et al.*, 2010). From these calculated RMSDs of overall structures, we observe the strength of the N3O structural motif in maintaining active site structure throughout the optimization process, as well as the interspecies sequence homology.

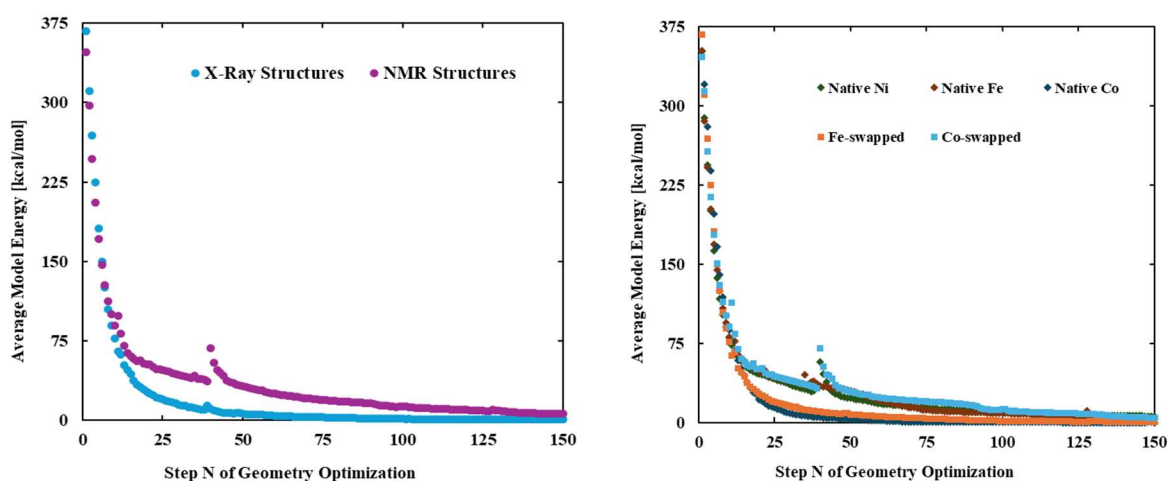


Fig 9. Model Energy per Step N of Normal SCF convergences in ORCA illustrating minimal metal-dependence of optimization, but large variance between workflows.

To ensure that our ensemble RMSD technique would be representative and unskewed by outliers within the ensemble, we split the protein by side chain conservation, calculating the arithmetic mean RMSD for the cofactor-ligating and substrate-binding residues (Table 4). By contextualizing our calculated numbers as a numeric measure of positional divergence, we can now conceptualize the full depth of arginine's potential role in reactivity, whose positioning shows the greatest divergence between the original and optimized structures across the board. Clearly, these residues should be conserved or at least considered in all future computational simulations of the active site. In simulations performed on 2HJI and Co-2HJI, Arginine 154 was positioned such that it could even bond the labile waters bound to the iron and cobalt, facilitating

their removal from the metal as hypothesized in all recent hypotheses of ARD reactivity (Fig. 10).

Table 4. Arithmetic means of RMSD values for each non-backbone residue included in the calculations, as well as their sum.

PDB ID	His'88/96	His'90/98	His'133/140	Glu'94/102	Arg'96/104	Arg'147/154	2(H ₂ O)M	Sum Δ
1ZRR	0.1413	0.2894	0.2709	0.2917	0.7400	2.677	0.7513	5.1618
Co-1ZRR	0.1403	0.3261	0.2730	0.2367	0.7400	2.757	0.7513	5.2245
2HJI	0.2418	0.3448	0.3328	0.3778	0.6700	2.313	1.048	5.3284
Co-2HJI	0.2625	0.3443	0.3026	0.5244	0.6512	2.217	1.099	5.4018
5I8S	0.0947	0.1274	0.1666	0.2283	0.6512	0.5654	0.6360	2.4696
Fe-5I8S	0.0949	0.2101	0.3145	0.2253	0.5107	0.9564	0.7358	3.0476
5I8Y	0.1653	0.2375	0.3429	0.4839	0.5667	0.6264	0.6360	3.0586
Fe-5I8Y	0.1683	0.2011	0.3173	0.4518	0.6298	0.7867	0.5666	3.1216

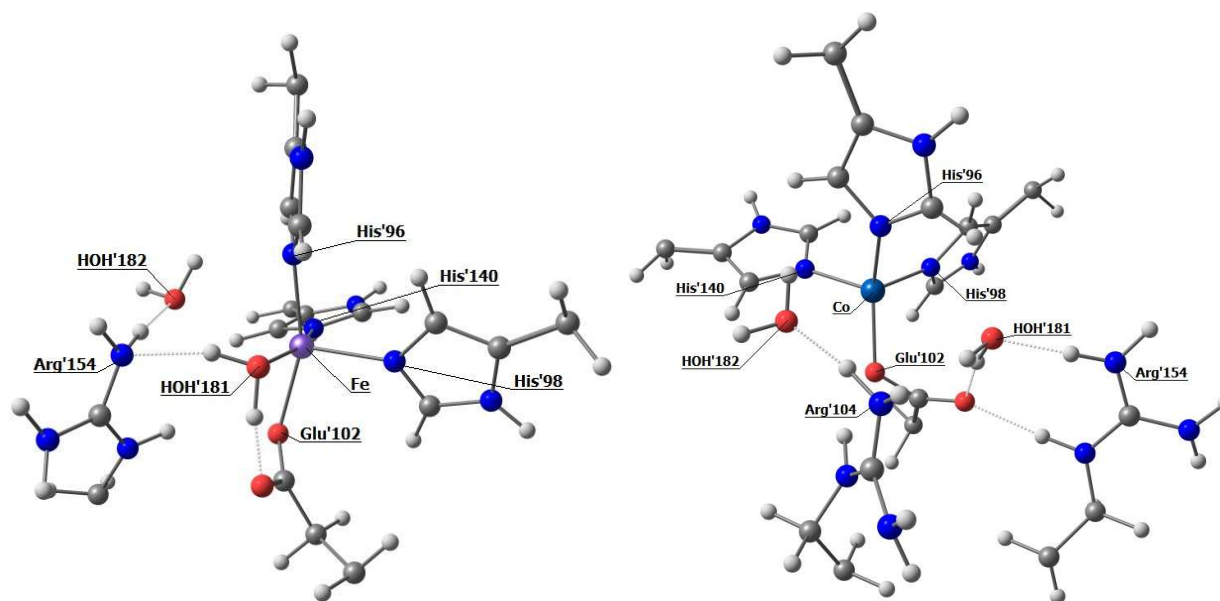


Fig 11. Zoomed in visualization of water molecule displacement in **2HJI** and **Co-2HJI** from their metal cofactors and onto arginine 154.

Although these geometry optimizations are no guarantee of any reaction pathway, these simulations of 2HJI and Co-2HJI provide some evidence for the arginine residue's role in the

removal and replacement of water in the absence of substrate, as well as Sparta *et al.*'s hypothesized deprotonation and stabilization of acireductone (Fig 11).

Also relevant to future considerations is that in the absence of both arginine 147's adjacent tryptophan 155 residue and the substrate, arginine 147/154 hydrogen bonds with its own backbone carbonyl. We report that the inclusion of tryptophan 155 in X-ray structures enables pi-stacking with the guanidino group of arginine 147/154, stabilizing it in an orientation relevant to the substrate's position in the active site (Fig 11). This inclusion further informed our caution regarding the inclusion of side chains in our models, leading us to include two phenylalanine groups, similarly conserved in the two large turn-chains that comprise the ARD active site. Visual analysis of our X-Ray models further reiterates that although there is little to no observed metal dependence in optimization (Fig 12), the workflow of the structure solution has a large influence on structural evolution (Table 5; Fig 13), corroborating our findings from the SCF convergences in optimization (Fig 8).

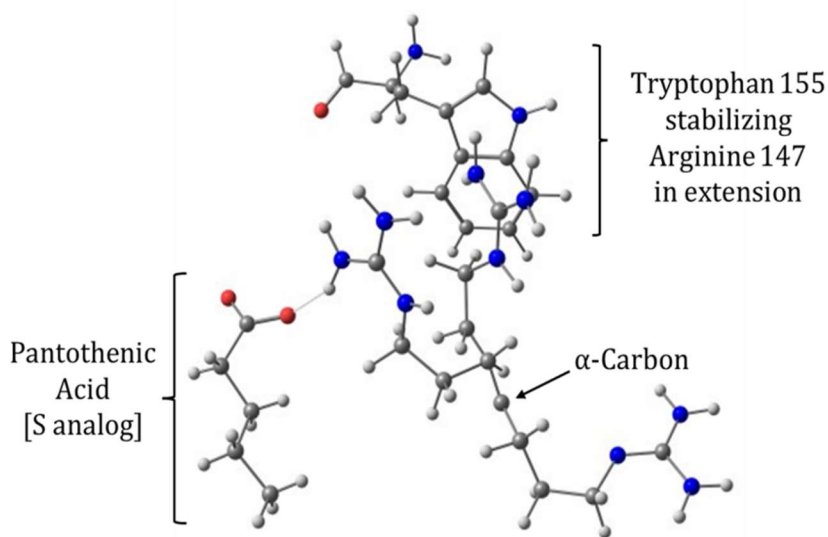


Fig 12. Illustration of Arg 147's optimized position (L-R) in the presence of a substrate analog, in the presence of Trp155, and in the absence of neither, modeled with Chemcraft.

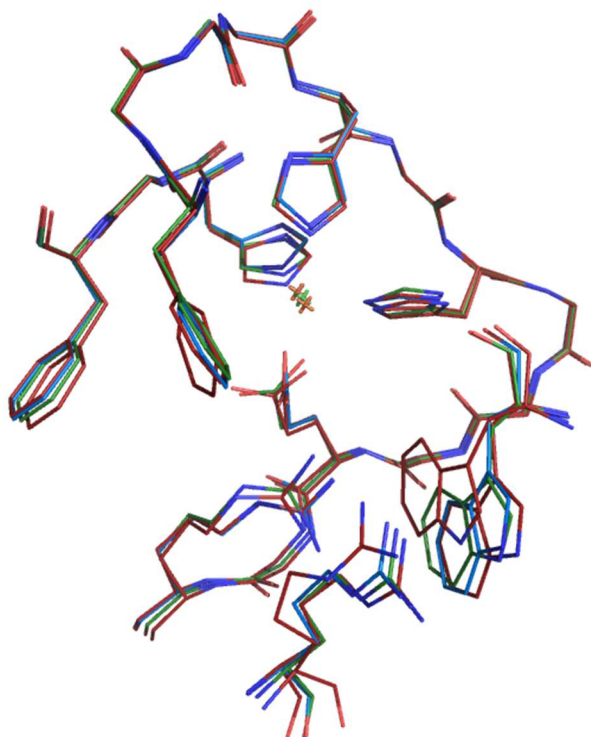


Fig 13. Optimized geometries of X-Ray solved Mm-ARD (**Ni**, **Co** and **Fe**), modeled with PyMol

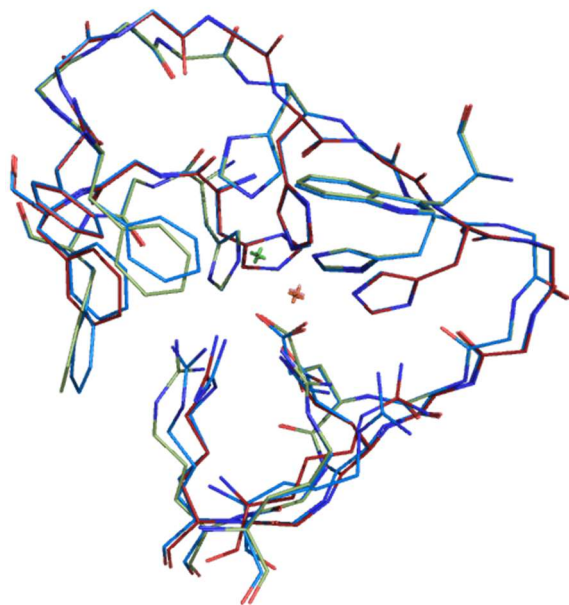


Fig 14. Optimized geometries of NMR-solved Ko-ARD (**Ni**, **Co** and **Fe**), modeled with PyMol

Ensemble RMSDs were also assembled by averaging a matrix of RMSDs between models along the diagonal, enabling us to quantitatively visualize overall differences between models and betraying no distinct metal-dependent structural similarities in the optimized structure (Table 5).

Table 5. Overall structural RMSD values between each optimized structure calculated using PyMol without outlier rejection, as well as the Ensemble RMSD.

Pymol Net RMSD of Optimized X-Ray/NMR Structures (Heavy Atoms, No Outlier Rejection)								
PDB ID(s)	1ZRR	Co-1ZRR	2HJI	Co-2HJI	5I8S	Fe-5I8S	5I8Y	Fe-5I8Y
1ZRR	0.000	0.301	4.090	4.067	6.551	6.510	6.585	6.604
Co-1ZRR	0.301	0.000	4.058	4.020	6.574	6.510	6.559	6.602
2HJI	4.090	4.058	0.000	0.457	4.383	4.355	4.503	4.438
Co-2HJI	4.067	4.020	0.457	0.000	4.289	4.287	4.394	4.363
5I8S	6.551	6.574	4.383	4.289	0.000	0.669	0.359	0.333
Fe-5I8S	6.510	6.510	4.355	4.287	0.669	0.000	0.823	0.882
5I8Y	6.585	6.559	4.503	4.394	0.359	0.823	0.000	0.206
Fe-5I8Y	6.604	6.602	4.438	4.363	0.333	0.882	0.206	0
Ensemble RMSD =		3.849	Angstroms					

A critical step in ensuring the validity of our models is the computation of the single-point energies and spin populations for each model, calculated with b3lyp, to better ensure that our models maintain some semblance of their original active site environment. The greatest challenge here was the sheer size of the Large Print models, each of them over a gigabyte and thus unloading in Chemcraft for examination. For this reason, we wrote a small Python script to parse each file and extract the “Total energy after final integration (Eh)” and the spin populations of the metals and their active sites (Table 6).

Table 6. Calculated single point energies, S^2 , optimized spin multiplicities and the spin populations of the metal and whole active site expressed as a percentage of total spin

SPEs & Spin Populations of the Overall Models & Active Sites						
PDB ID	SPE [Eh]	$\langle S^2 \rangle$	2S + 1	M(II)	Active Site	% Retained
1ZRR	-8052.9525	2.00	3.00	1.68	1.99	99.5%
Co-1ZRR	-7927.3916	3.76	4.00	2.75	2.96	98.5%
2HJI	-7197.8210	6.01	5.00	3.81	3.91	97.8%
Co-2HJI	-7316.8852	3.76	4.00	2.74	2.89	96.3%
5I8S	-8052.9685	2.00	3.00	1.67	1.99	99.4%
Fe-5I8S	-7197.8210	6.01	5.00	3.81	3.91	97.8%
5I8Y	-7927.4203	3.76	4.00	2.75	2.95	98.3%
Fe-5I8Y	-7808.3652	6.01	5.00	3.80	3.93	98.3%

Although there is inconsistent evidence for metal dependence visible in the single point energies of our models, namely in Co-2HJI, we report strong conservation of the initial input spin states of the models despite drastic variance in optimized structure, the metals seem to maintain much of their electronic environment. This observation is corroborated by our expanded parsing of the output file, examining the spin populations and Mulliken atomic charges of the metals and their ligating atoms (Table 7, 8). The seemingly metal dependent overlaps in both spin and charge, especially in Fe-ARD (2HJI and Fe-5I8S in particular), point to some potential parallels in reactivity, despite their demonstrated structural difference (with a full 4.355 RMSD between them). Of note, here is a refutation of the previously hypothesized Lewis acid character of nickel, where its spin population is noticeably more spread between the surrounding active site across all our models, whereas iron is more stoic. Here, cobalt is skewed from the middle, and more towards nickel as its reactivity in ARD would suggest.

Although Mulliken atomic charges are notoriously suboptimal in DFT calculations of transition metal complexes, spins should be modeled accurately in b31yp, so although there

should be some reasonable doubt towards our representations of charge, our spin populations and single point energy comparisons should be valid and certainly warrant further investigation.

Table 7. Calculated DFT spin populations of metals & ligating atoms

Optimized Spin Populations of M(II) and Ligating Atoms in X-Ray/NMR Structures							
PDB ID	M(II)	His'88/96	His'90/98	His'133/140	Glu'94/102	HOH'311/181	HOH'323/182
1ZRR	1.68	0.07	0.08	0.07	0.03	0.04	0.03
Co-1ZRR	2.75	0.04	0.05	0.05	0.03	0.03	0.02
2HJI	3.81	0.02	0.03	0.02	0.01	0.03	0.00
Co-2HJI	2.74	0.04	0.05	0.06	0.01	0.00	0.00
5I8S	1.67	0.07	0.07	0.08	0.04	0.04	0.03
Fe-5I8S	3.81	0.02	0.03	0.02	0.01	0.03	0.00
5I8Y	2.75	0.04	0.05	0.04	0.03	0.02	0.03
Fe-5I8Y	3.80	0.02	0.03	0.03	0.02	0.02	0.02

Table 8. Calculated Mulliken atomic charges of metals & ligating atoms

Optimized Mulliken Atomic Charges of M(II) and Ligating Atoms							
PDB ID	M(II)	His'88/96	His'90/98	His'133/140	Glu'94/102	HOH'311/181	HOH'323/182
1ZRR	0.26	-0.26	-0.25	-0.28	-0.61	-0.28	-0.39
Co-1ZRR	0.35	-0.31	-0.29	-0.33	-0.53	-0.31	-0.41
2HJI	0.58	-0.39	-0.39	-0.33	-0.61	-0.34	-0.33
Co-2HJI	0.50	-0.39	-0.30	-0.31	-0.55	-0.42	-0.34
5I8S	0.23	-0.26	-0.29	-0.25	-0.48	-0.30	-0.22
Fe-5I8S	0.58	-0.39	-0.39	-0.33	-0.61	-0.34	-0.33
5I8Y	0.37	-0.30	-0.30	-0.29	-0.45	-0.30	-0.29
Fe-5I8Y	0.51	-0.36	-0.33	-0.32	-0.46	-0.31	-0.31

Conclusions and Future Directions

Our models indicate no quantitative metal dependence in the overall optimized ARD structure in the absence of substrate, suggesting that the metal dependence of reactivity likely arises from other parameters, such as the evolution of structural differences after the binding of acireductone or charge transfer differences between the metals. Future research should focus on reintroducing substrate into these models, as DFT can model reactivity to a limited extent, and interrogating spin population differences between optimized models and hypothesized transition

states. Recent work on small-model recreations of reactivity should also be tested and expanded upon, as on-pathway chemistry has been reproduced in ferrous mononuclear simulations of the active site. Altogether, this work represents a step forward in simplified DFT simulations of protein native states and in quantitative measurements of univariate isozyme alignment.

Appendix A – Annotated ORCA Slurm File

#!/bin/bash

#SBATCH --job-name parallel-orca

#SBATCH --ntasks=8

#SBATCH --cpus-per-task=1

#SBATCH --mem=32G

#SBATCH --output=OutputFileName.out

#SBATCH --partition=r6615

#SBATCH --time=48:00:00

cd ~/InputDirectory

/home/software/orca/orca.sif InputFileName.inp

Appendix B – Annotated ORCA Input File for Geometry Optimizations

```
# Species, M2+-bound, PDB ID (ARD), Structure. w/ M2+ ox state, normal optimization
#Number of atoms,
#List of fully included residues
#List of abbreviated
! bp86 def2-tzvp pal8
! kdiis slowconv opt d3bj normalprint normalscf

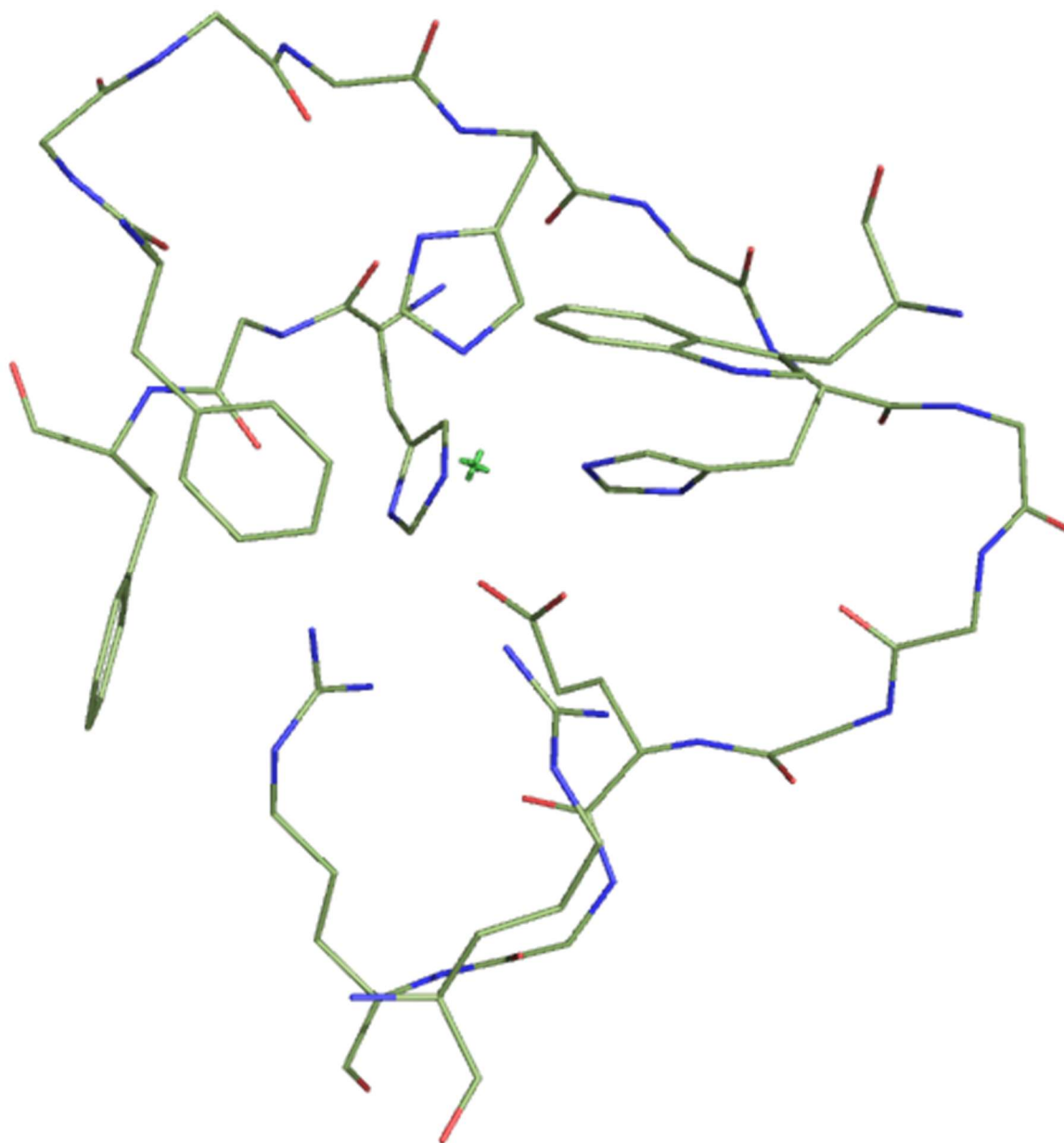
% basis
    newgto H "def2-svp" end
    newgto C "def2-svp" end
end

%scf
    maxiter 512
end

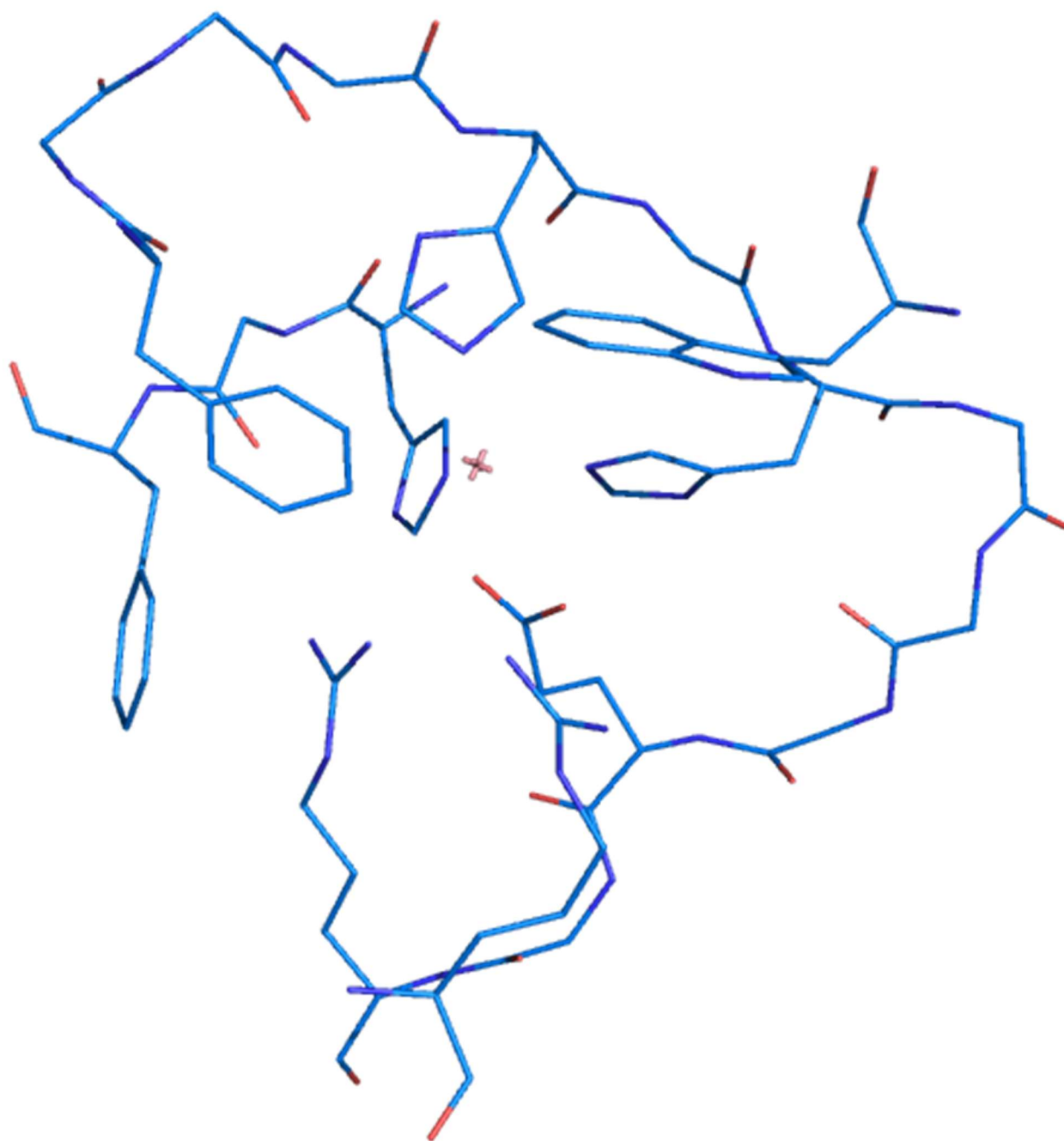
# update the geom block for each model
%geom
    constraints #Alpha Carbon and Hydrogen
        #{C Alpha Carbon or Hydrogen Atom Number C}
    end
end

# update charge (a), spin multiplicity (b) for each model
*xyz a b
#List xyz coordinates
*
```

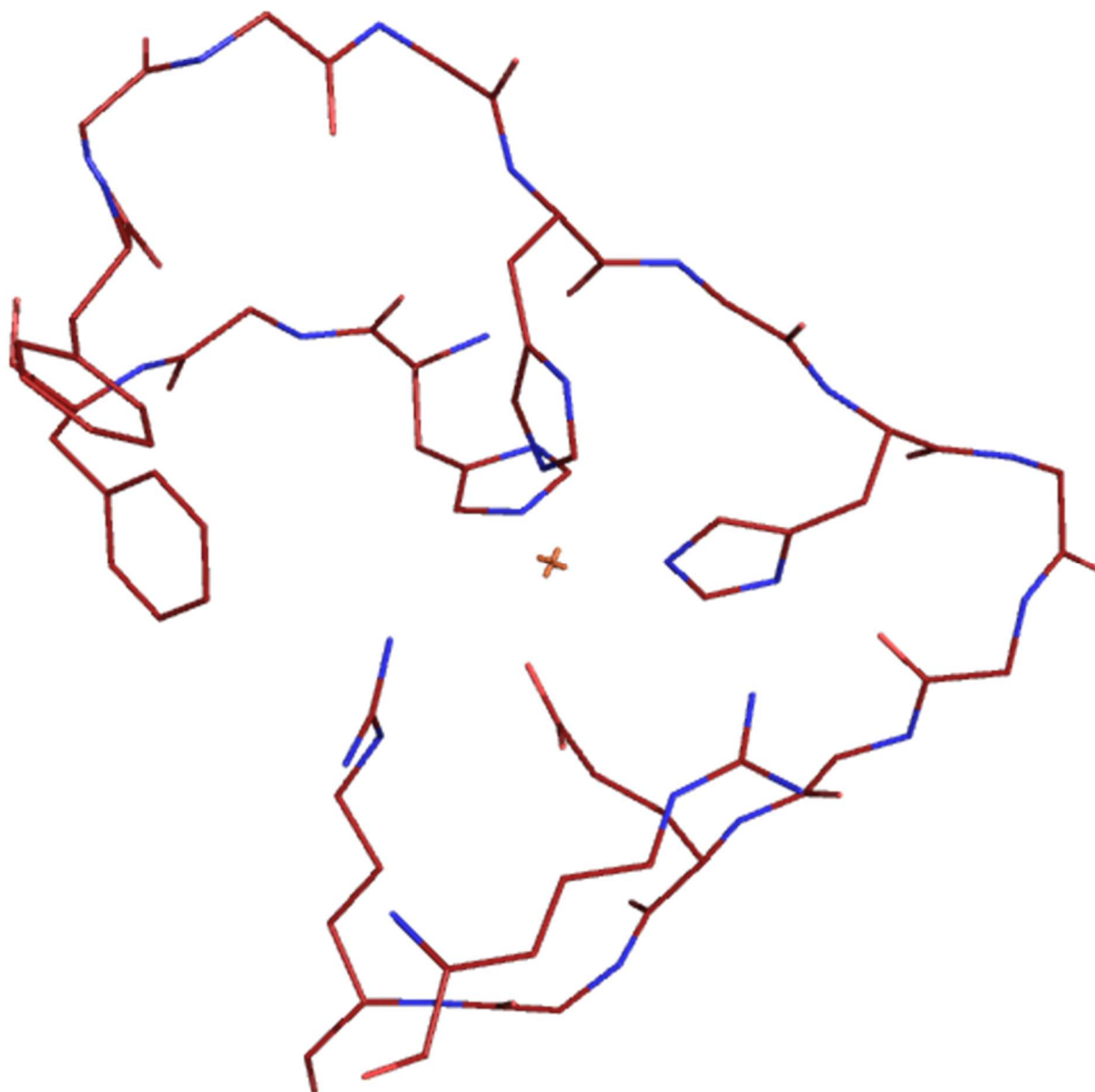
Appendix C – Individual Structural Models, Modeled in PyMol



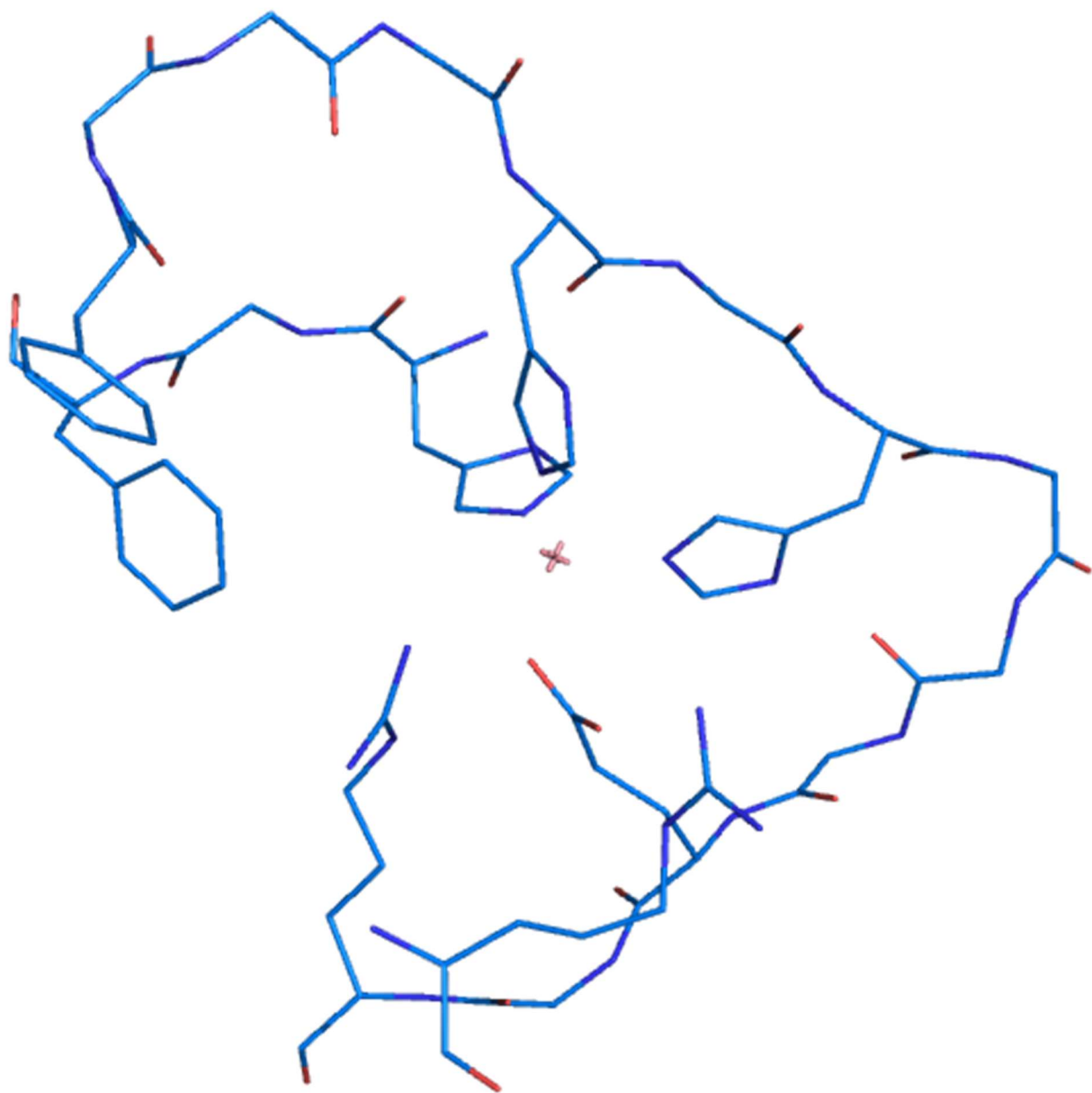
Heavy Atom Structure of Solvent Optimized Geometry of 1ZRR



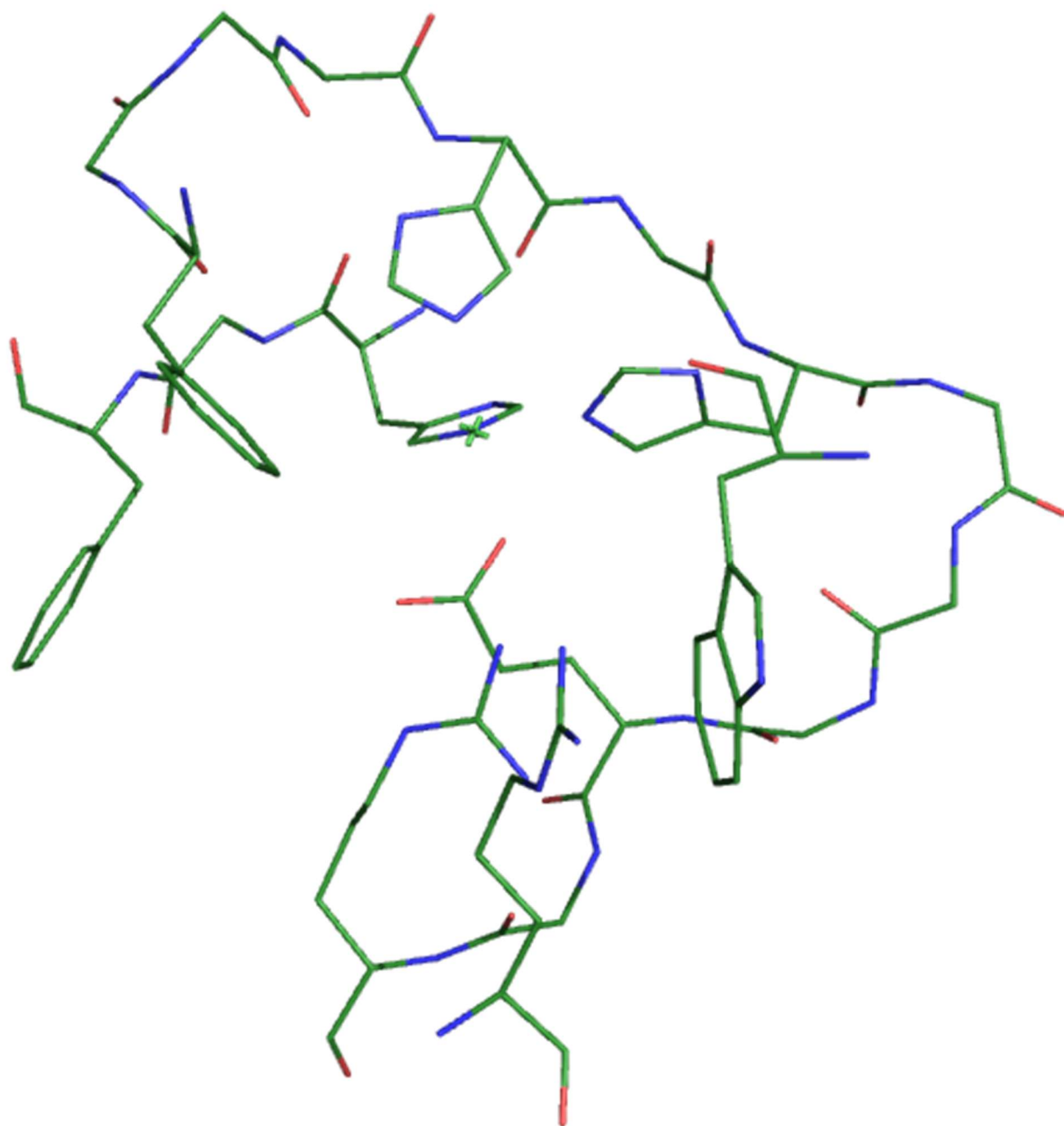
Heavy Atom Structure of Solvent Optimized Geometry of Co1ZRR



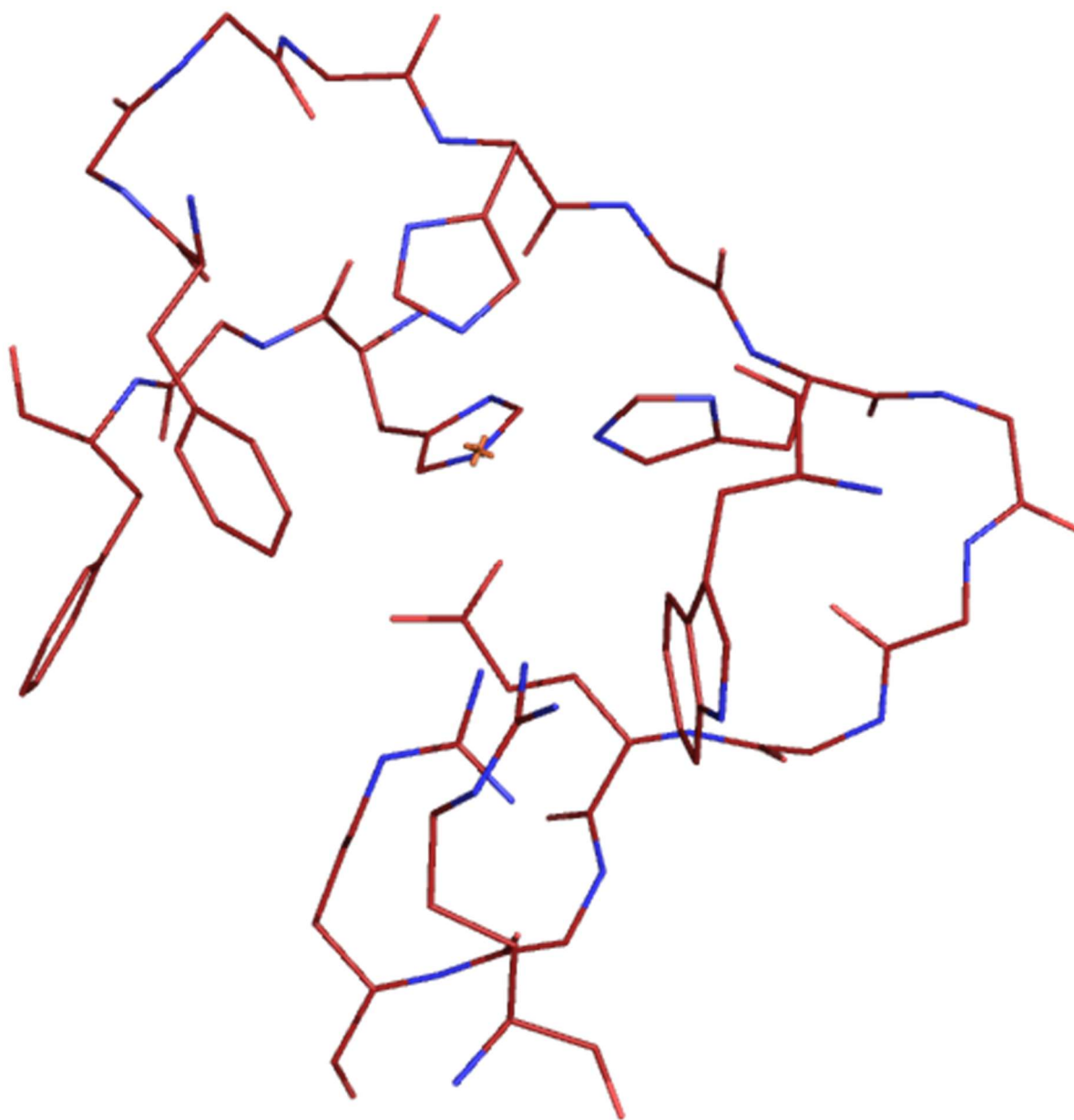
Heavy Atom Structure of Solvent Optimized Geometry of 2HJI



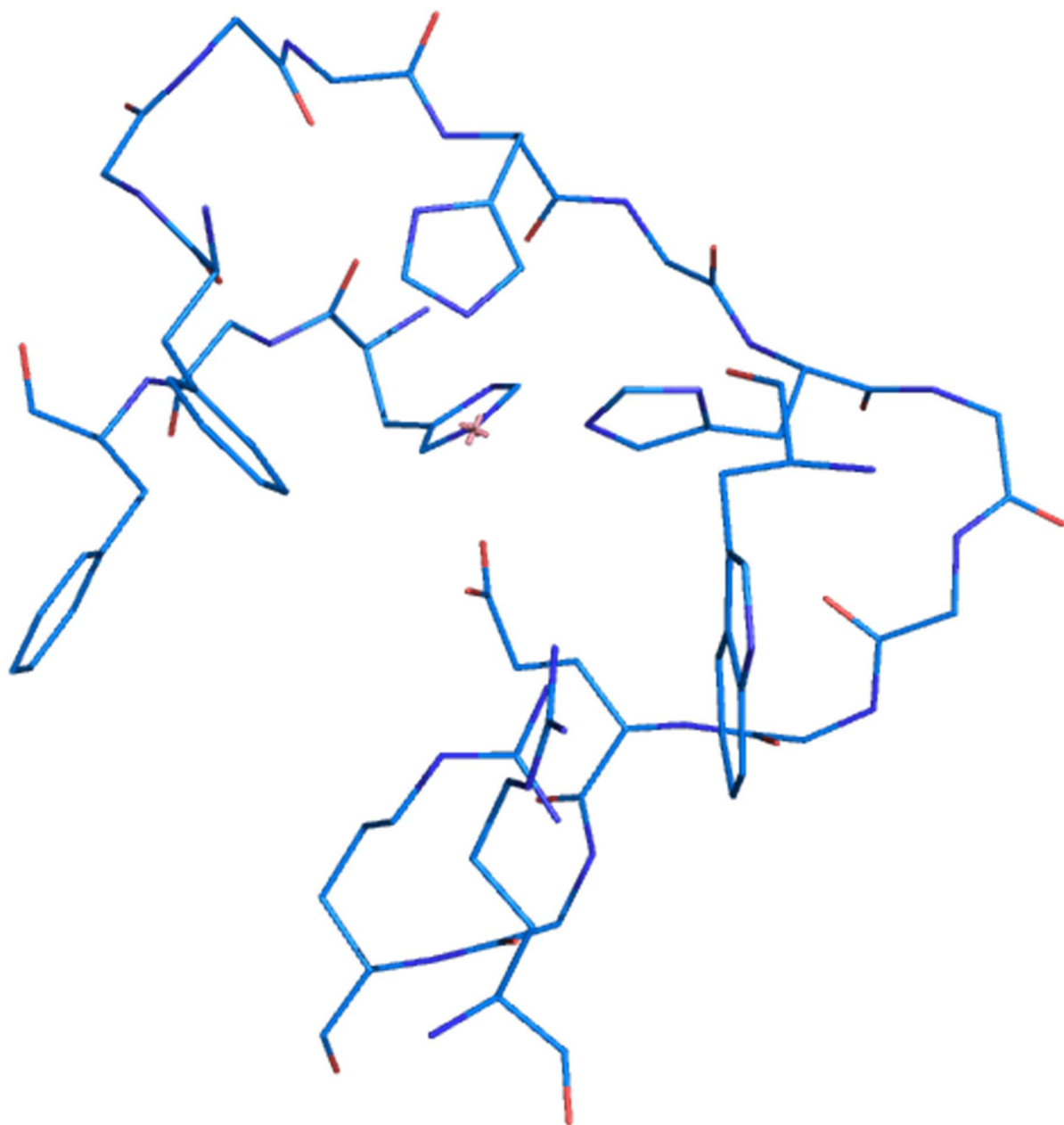
Heavy Atom Structure of Solvent Optimized Geometry of Co2HJI



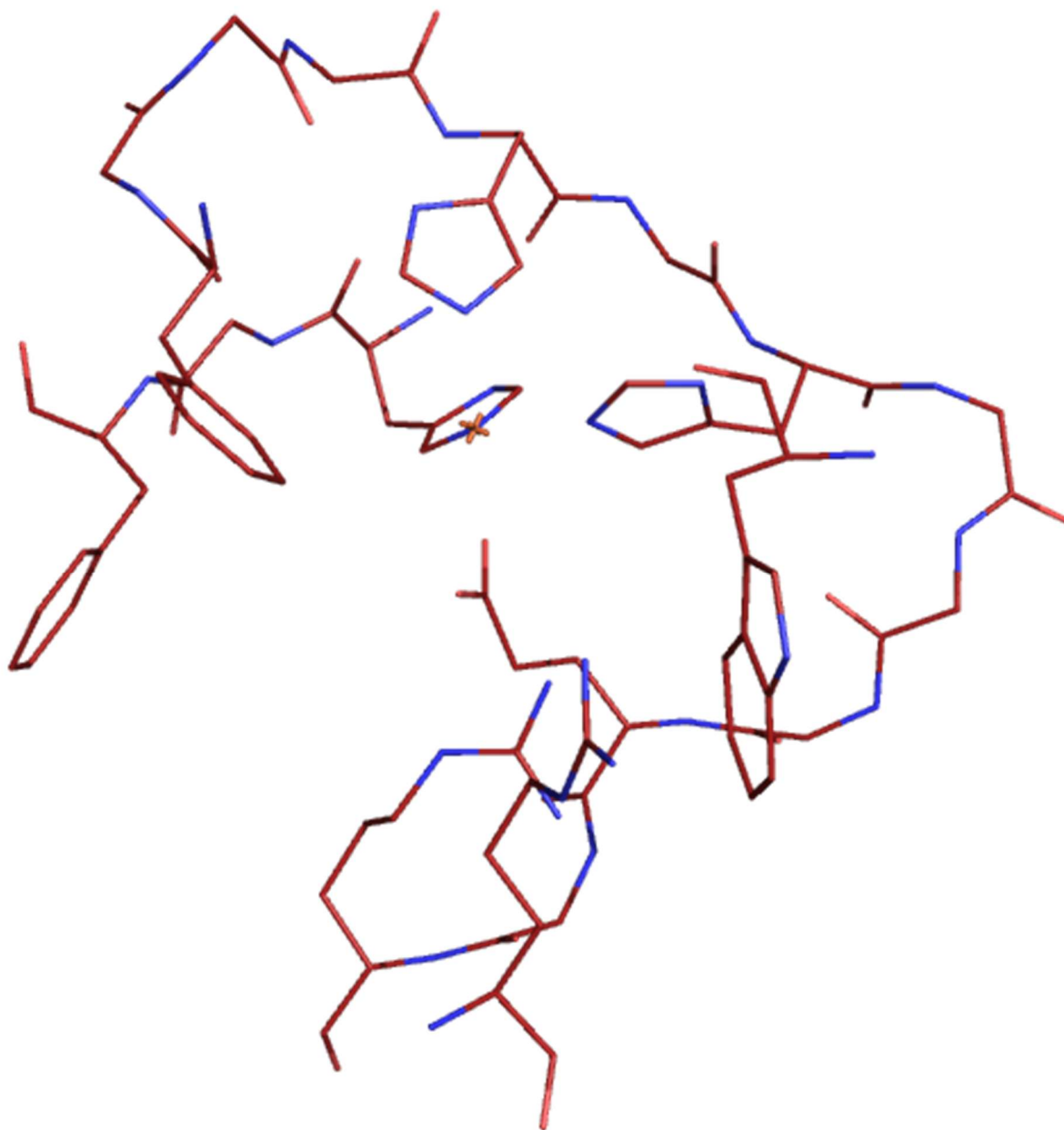
Heavy Atom Structure of Solvent Optimized Geometry of 5I8S



Heavy Atom Structure of Solvent Optimized Geometry of Fe5I8S



Heavy Atom Structure of Solvent Optimized Geometry of 5I8Y



Heavy Atom Structure of Solvent Optimized Geometry of Fe5I8Y

Appendix D: Annotated LargePrint Input File for SPE Calculations

```
# Species, M2+-bound, PDB ID (ARD), Strucure. w/ M2+ ox state, normal optimization
#Number of atoms,
#List of fully included residues
#List of abbreviated
#single point energy calc. using b3lyp
! b3lyp def2-tzvp pal8 cpcm largeprint tightscf

% basis
    newgto H "def2-svp" end
    newgto C "def2-svp" end
end

%scf
    maxiter 512
end

%cpcm
    epsilon 4
end

# update charge (a) and spin multiplicity (b) for each model (Atom Count)
*xyz 3 3
#xyz coordinates

*
```

Appendix E – Annotated LargePrint Parser for SPE files

```
#Run just once to build the dataframe, which will then house all necessary information
!pip install orca_parser
import orca_parser
import re
import pandas as pd
energies = pd.DataFrame(columns=['File Path', 'Total energy after final integration (Eh)'])
```

```
#Function to check the last 7 lines in the file to ensure ****ORCA COMPLETED
NORMALLY****
```

```
def failureCheck(file_path):
    print("\n--- End of File ---")
    # Read all lines and print the last 7 lines
    with open(file_path, 'r') as f:
        lines = f.readlines()
        for line in lines[-7:]:
            print(line.strip())
```

```
#Function to parse for all lines featuring Energy and return the Last Iteration Total Energy
```

```
def allElines(file_path):
    print(f'Searching for relevant energy lines in {file_path}:')
    energies_found = {}
    with open(file_path, 'r') as f:
        for line in f:
            #if 'energy' in line.lower():
```

```

# print(f'DEBUG: Found 'energy' in line (repr): {repr(line)}')
# print(f'DEBUG: Found 'energy' in line (stripped): {line.strip()}')

stripped_line = line.strip()
if "Total energy after final integration" in stripped_line:
    match = re.search(r'Total energy after final integration\s*:\s*(-?\d+\.\d+)', stripped_line)
    if match:
        energies_found['Total energy after final integration'] = float(match.group(1))
elif "Last Iteration Total Energy:" in stripped_line:
    match = re.search(r'Last Iteration Total Energy:\s*(-?\d+\.\d+)', stripped_line)
    if match:
        energies_found['Last Iteration Total Energy'] = float(match.group(1))
return energies_found

```

#Function to setup data for entry into the dataframe and return Spin Populations and Mulliken Atomic Charges

```

def makeATable(file_path, energy_components, spins_dict, charges_dict):
    final_integration_energy = energy_components.get('Total energy after final integration',
    energy_components.get('Last Iteration Total Energy'))
    data = {
        'File Path': [file_path],
        'Total energy after final integration (Eh)': [final_integration_energy]
    }
    # Add each spin population as a separate column
    for atom_id, spin_val in spins_dict.items():
        data[f'S_{atom_id}'] = [spin_val]
    # Add each charge as a separate column
    for atom_id, charge_val in charges_dict.items():

```

```
data[f'C_{atom_id}'] = [charge_val]
return data
```

```
#Function to extract all Mulliken Atomic Charges and Spin Populations
```

```
def extract_mulliken_data(file_path, atom_identifiers):
```

```
    spins_found = {atom_id: None for atom_id in atom_identifiers} # Initialize with None
```

```
    charges_found = {atom_id: None for atom_id in atom_identifiers} # Initialize with None
```

```
    print(f'Searching for mulliken data in '{file_path}' for atoms: {atom_identifiers}')
```

```
    try:
```

```
        with open(file_path, 'r') as f:
```

```
            lines = f.readlines()
```

```
    except FileNotFoundError:
```

```
        print(f'Error: File not found at {file_path}')
```

```
        return spins_found, charges_found
```

```
mulliken_section_lines = []
```

```
found_mulliken_start = False
```

```
for i, line in enumerate(lines):
```

```
    stripped_line = line.strip()
```

```
    if 'MULLIKEN ATOMIC CHARGES AND SPIN POPULATIONS' in stripped_line:
```

```
        found_mulliken_start = True
```

```
        print(f'Entered MULLIKEN ATOMIC CHARGES AND SPIN POPULATIONS  
section.')
```

```
        # Skip header lines: 'CHARGE', 'SPIN', and separator lines
```

```
        # and the section header itself. Assuming these are 3 lines after the header itself.
```

```

# Start gathering actual data lines from i+4
mulliken_section_lines = lines[i+4:]
break # Found the section, now process it

if not found_mulliken_start:
    print(f'Warning: 'MULLIKEN ATOMIC CHARGES AND SPIN POPULATIONS' section
not found in '{file_path}'.")
    return spins_found, charges_found

# Process all lines in the Mulliken section to find all requested spins and charges
for section_line in mulliken_section_lines:
    stripped_section_line = section_line.strip()
    if not stripped_section_line or '-----' in stripped_section_line: # End of section or
separator
        break # End of Mulliken section data

parts = stripped_section_line.split()
if len(parts) >= 4: # Expected format minimum: Index Element: Charge Spin
    try:
        atom_index = parts[0]
        element_symbol = "
        charge_val = None
        spin_val = None

# Handle different formats for element symbol and colon
if ':' in parts[1]: # e.g., 'Fe:' as the second part
    element_symbol = parts[1].rstrip(':')
    if len(parts) >= 4:
        charge_val = float(parts[2])
        spin_val = float(parts[3])

```

```

elif len(parts) >= 5 and parts[2] == ':': # e.g., 'N', ':' as separate parts
    element_symbol = parts[1]
    charge_val = float(parts[3])
    spin_val = float(parts[4])
else: # Default to assuming element is parts[1] and charge/spin are parts[2] and parts[3]
    element_symbol = parts[1]
    if len(parts) >= 4:
        charge_val = float(parts[2])
        spin_val = float(parts[3])
    else:
        continue # Unrecognized format, skip this line

# Check if this line corresponds to any of the requested atom_identifiers
for atom_id in atom_identifiers:
    match_id = re.match(r'([A-Za-z]+)(\d+)?', atom_id)
    if not match_id:
        continue # Skip invalid atom_id format in the list

    requested_element = match_id.group(1)
    requested_index = match_id.group(2)

    if requested_index: # Specific atom by element and index, e.g., 'N11'
        if element_symbol == requested_element and atom_index == requested_index:
            if spins_found[atom_id] is None: # Only assign if not already found (handle
duplicates if any, though unlikely)
                spins_found[atom_id] = spin_val
                charges_found[atom_id] = charge_val
                break # Found for this atom_id, move to next atom_id in the inner loop
    else: # Only element symbol, e.g., 'Fe' or 'Co' (assuming first occurrence or single
such atom)

```

```

        if element_symbol == requested_element:
            if spins_found[atom_id] is None:
                spins_found[atom_id] = spin_val
                charges_found[atom_id] = charge_val
                break # Found for this atom_id, move to next atom_id in the inner loop

    except (IndexError, ValueError) as e:
        print(f'Parsing error on line: '{stripped_section_line}' - {e}')
        continue

# Report any atoms for which data was not found
for atom_id in atom_identifiers:
    if spins_found[atom_id] is None or charges_found[atom_id] is None:
        print(f'Warning: Mulliken data for '{atom_id}' not found in '{file_path}' or could not be
        parsed.")

return spins_found, charges_found

```

```

#List the atoms to look for Mulliken Atomic Charges and Spin Populations, set it to a variable
#e.g. 'Fe', 'Co', 'Ni', 'N9', 'N26', etc.
ATOMS_TO_TRACK = []

```

```

#Upload files one at a time into the session runtime
#Uncomment them each in sequence to save their file path to a variable
#Make sure to delete the files from the runtime after parsing then

```

```
variable1 = "file path 1"
#variable2 = "file path 2"
#variable3 = "file path 3"
```

```
#Change variables taken in each function to the corresponding one with the desired file path
energy_components = allElines(variable1)
failureCheck (variable1)
# Pass the list of atoms to track to extract_mulliken_data
spins_data, charges_data = extract_mulliken_data(variable1, ATOMS_TO_TRACK)
new_row_df = pd.DataFrame(makeATable(variable1, energy_components, spins_data,
charges_data))
energies = pd.concat([energies, new_row_df], ignore_index=True)
#The whole dataframe might not be shown, but it should be sanity checked to make sure that
ATOMS_TO_TRACK is correctly set
energies
```

```
#Sanity check each file, technically unnecessary but can be useful in deciding atoms to track
def findAllMullikenAndSpin(variable1):
    found_section = False
    header_line_count = 0

    with open(file_path, 'r') as f:
        for line in f:
            if 'MULLIKEN ATOMIC CHARGES AND SPIN POPULATIONS' in line:
                found_section = True
                print(line.strip()) # Print the header line
```

```
header_line_count = 0
```

```
continue
```

```
if found_section:
```

```
    # Print the 3 header lines (labels and separator) after the section title
```

```
    if header_line_count < 3:
```

```
        print(line.strip())
```

```
        header_line_count += 1
```

```
    else:
```

```
        # Print data lines until an empty line or a separator is found
```

```
        if not line.strip() or '-----' in line or 'ORCA' in line:
```

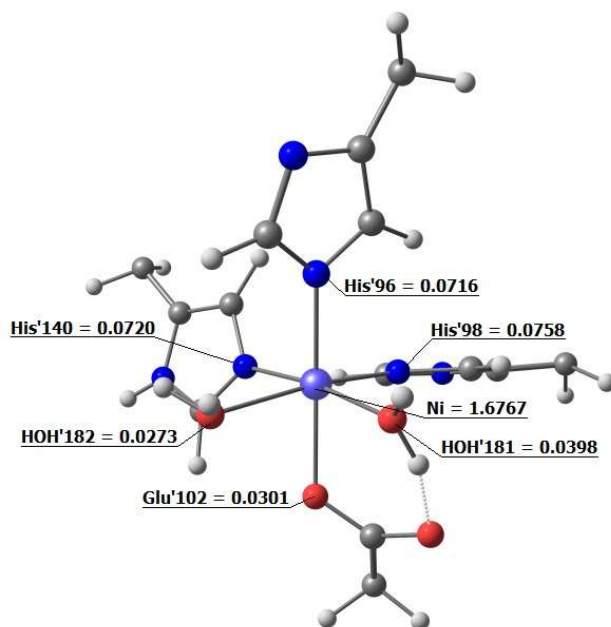
```
            break # End of section
```

```
        print(line.strip())
```

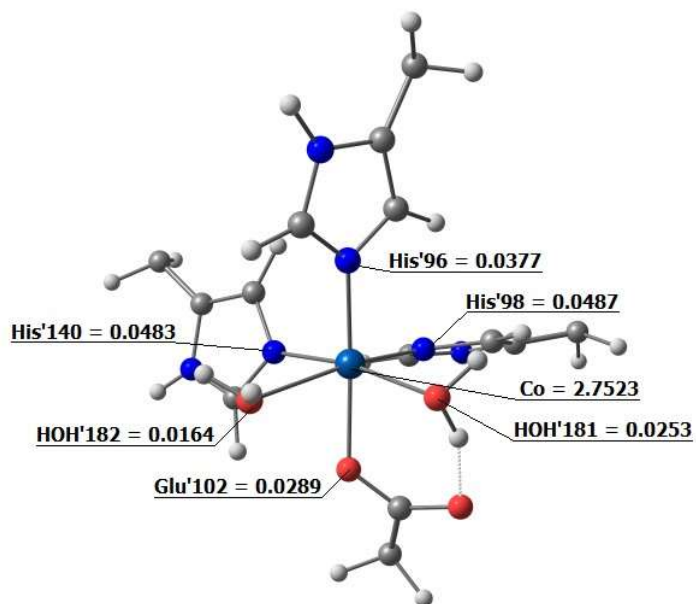
```
#Export the file to a .csv to load elsewhere (Excel)
```

```
energies.to_csv('energies_data.csv', index=False)
```

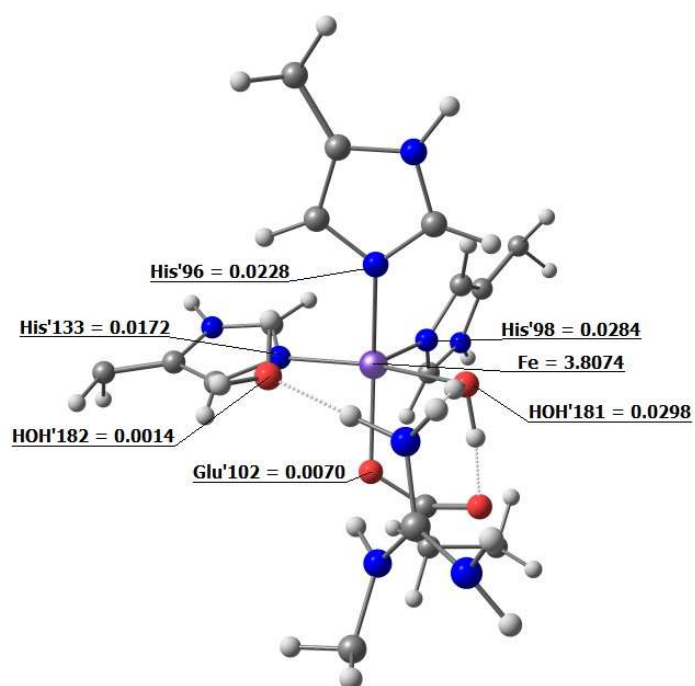
Appendix F – Map of Individual Active Site Spin Populations from LargePrint Parser



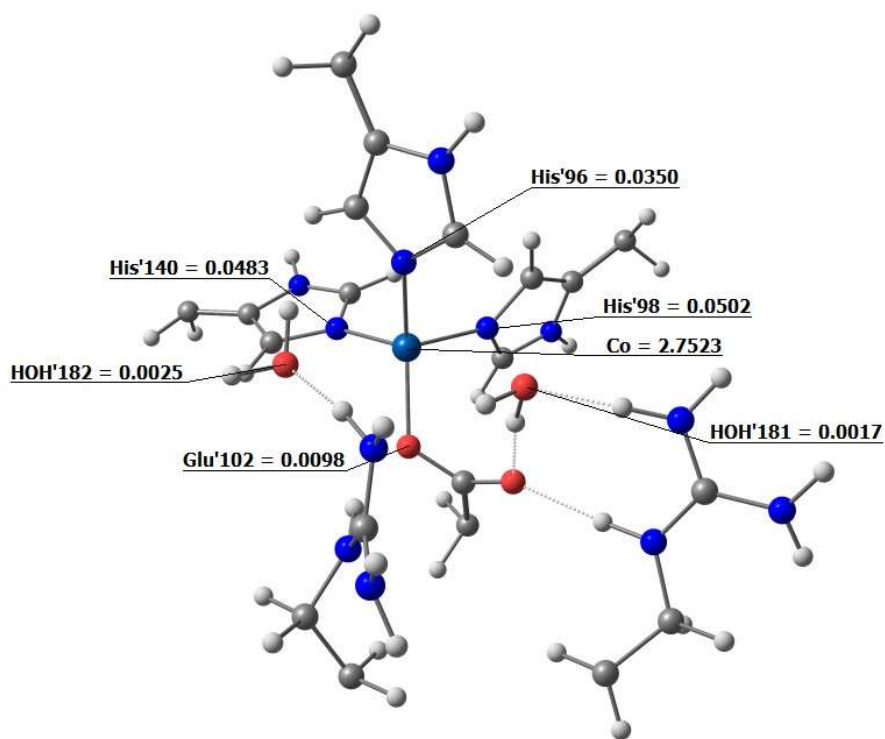
Active site map of spin populations of M and ligating atoms in 1ZRR.



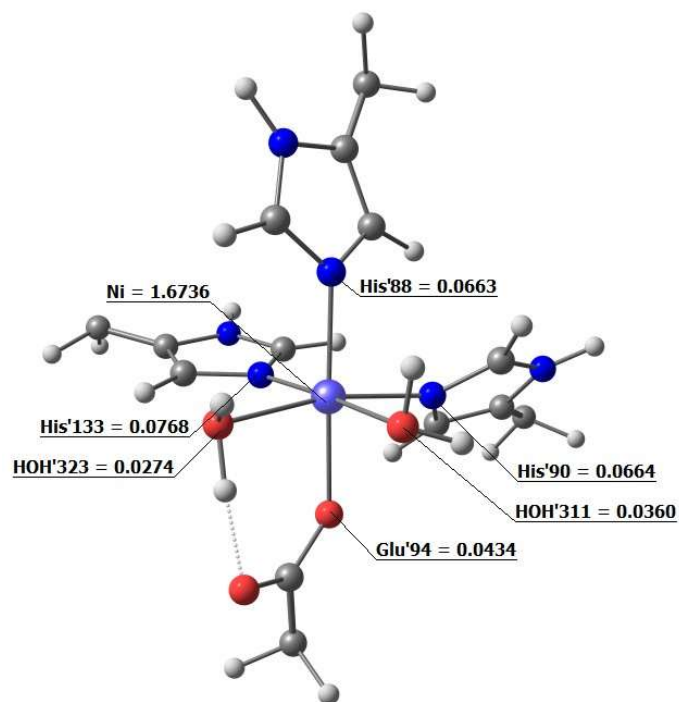
Active site map of spin populations of M and ligating atoms in Co-1ZRR.



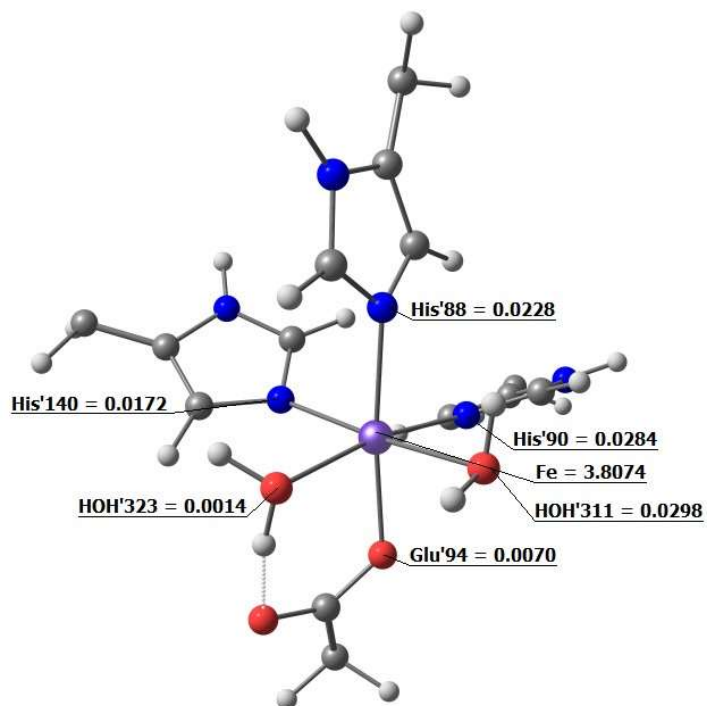
Active site map of spin populations of M and ligating atoms in 2HJI.



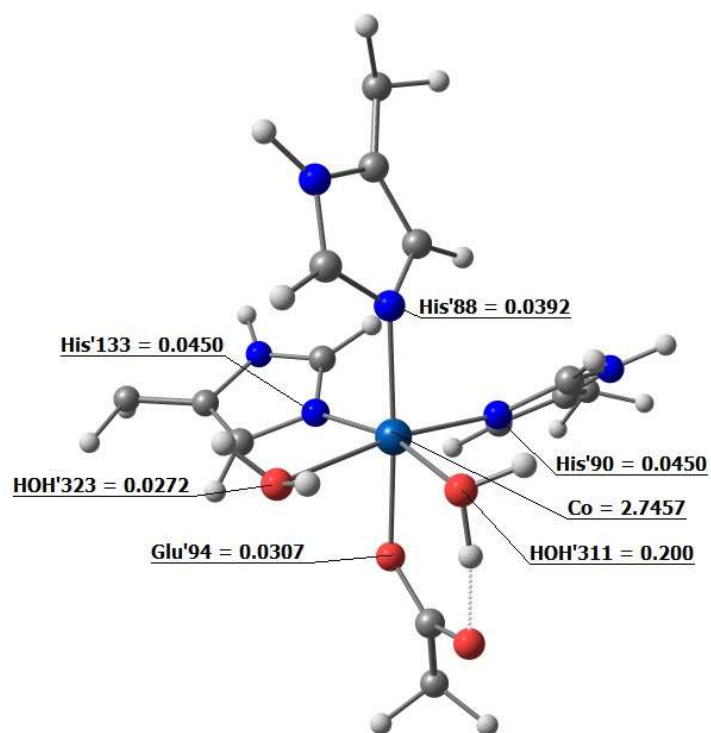
Active site map of spin populations of M and ligating atoms in Co-2HJI.



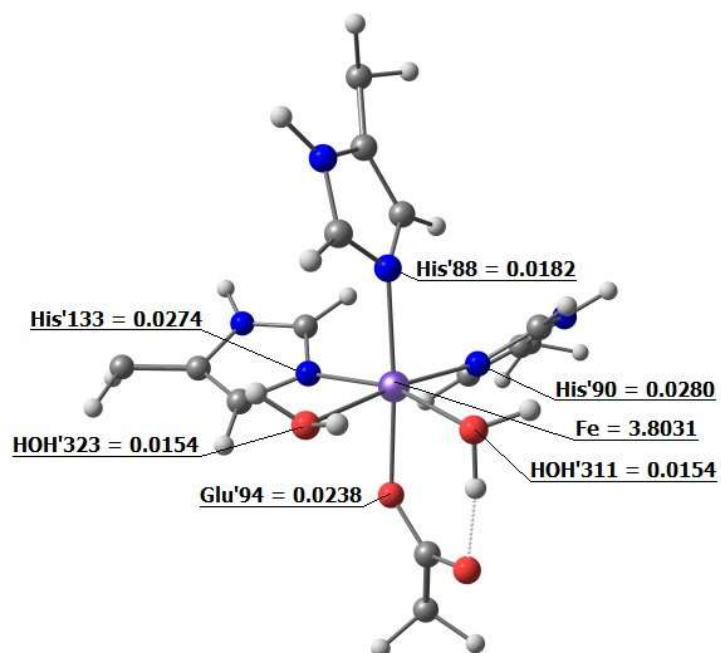
Active site map of spin populations of M and ligating atoms in 5I8S.



Active site map of spin populations of M and ligating atoms in Fe-5I8S.

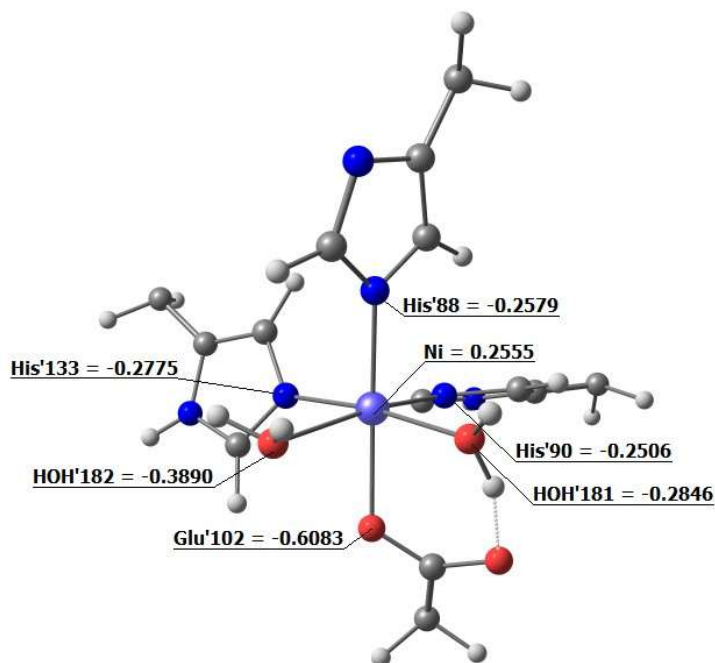


Active site map of spin populations of M and ligating atoms in 5I8Y.

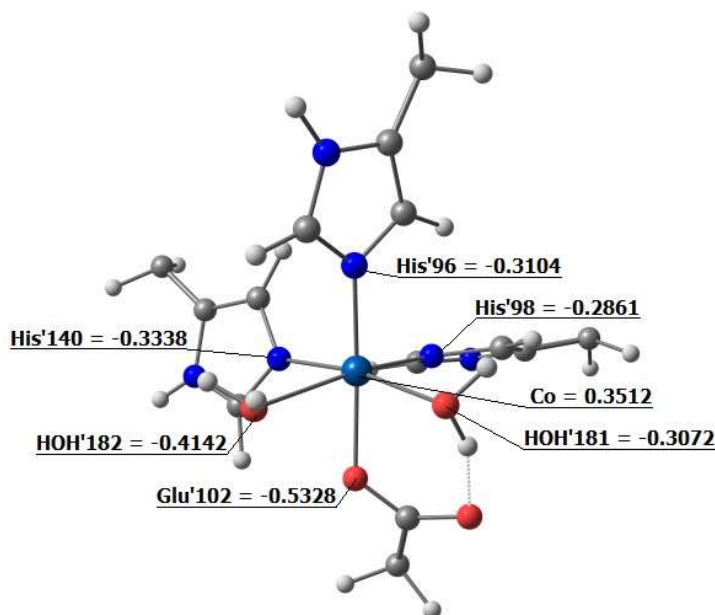


Active site map of spin populations of M and ligating atoms in Co-1ZRR.

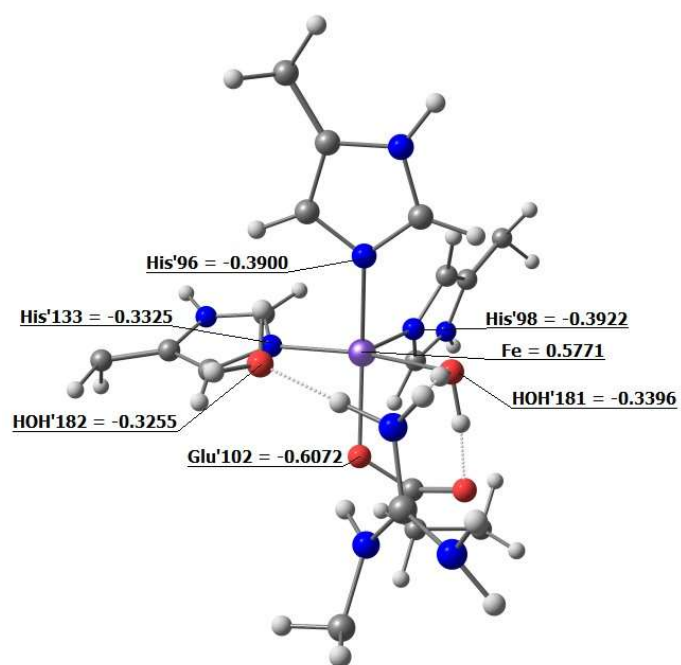
Appendix G – Map of Individual Active Mulliken Atomic Charges from LargePrint Parser



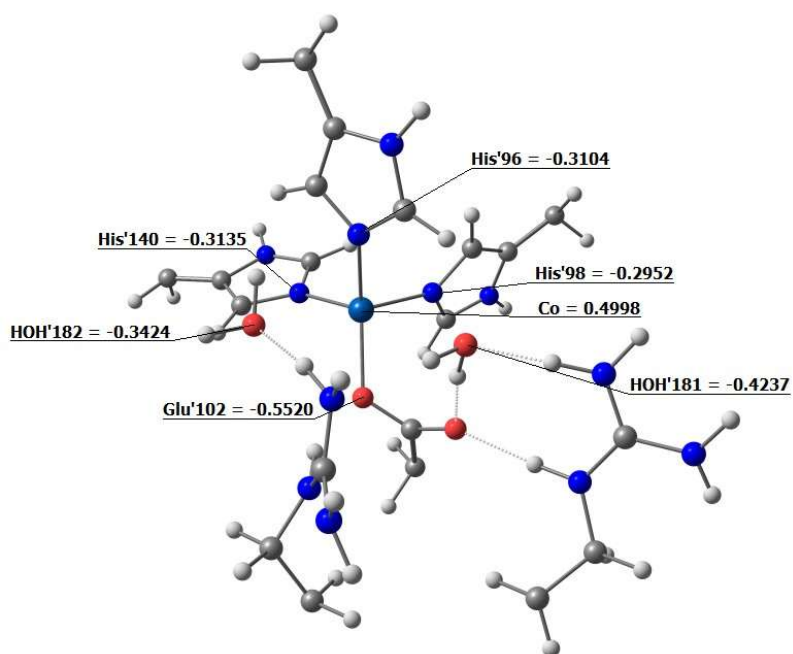
Active site map of Mulliken atomic charges of M and ligating atoms in 1ZRR.



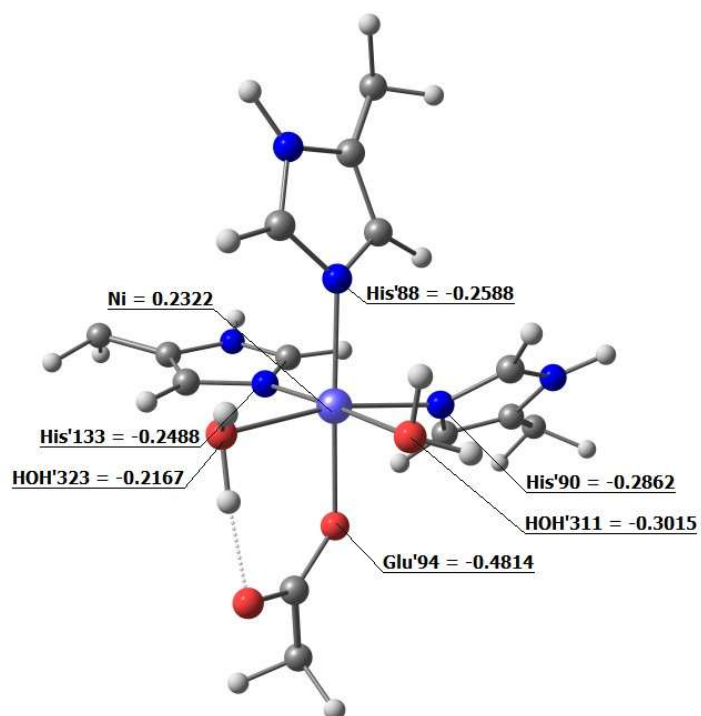
Active site map of Mulliken atomic charges of M and ligating atoms in Co-1ZRR.



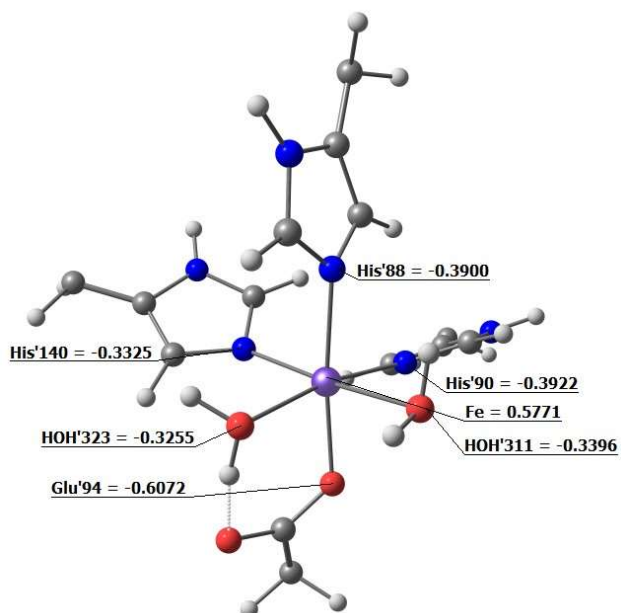
Active site map of Mulliken atomic charges of M and ligating atoms in 2HJI.



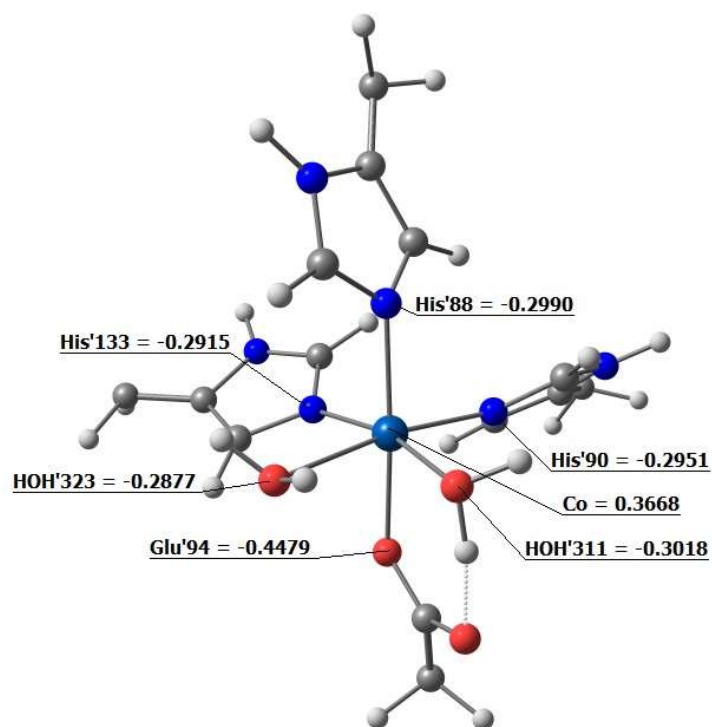
Active site map of Mulliken atomic charges of M and ligating atoms in Co-2HJI.



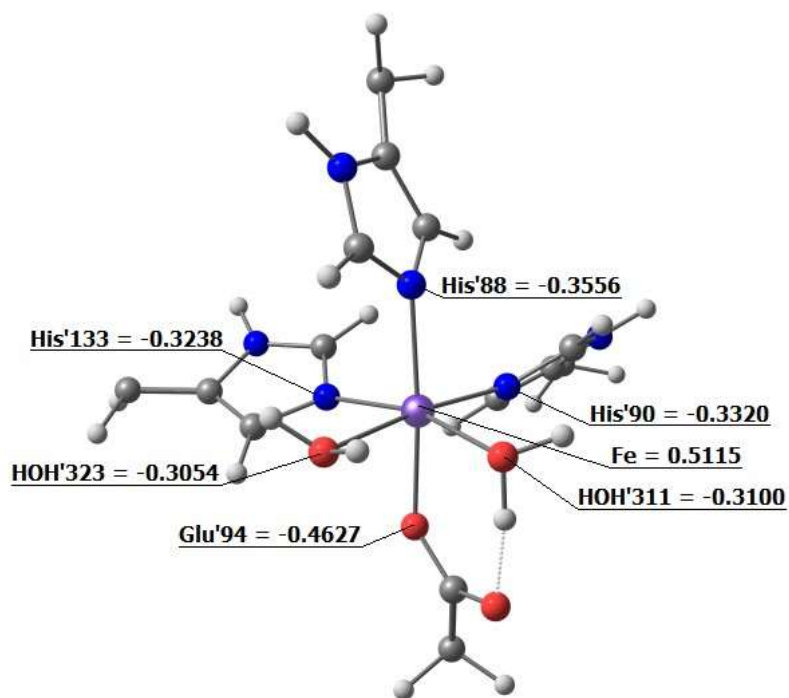
Active site map of Mulliken atomic charges of M and ligating atoms in 5I8S.



Active site map of Mulliken atomic charges of M and ligating atoms in Fe-5I8S.



Active site map of Mulliken atomic charges of M and ligating atoms in 5I8Y.



Active site map of Mulliken atomic charges of M and ligating atoms in Co-1ZRR.

Bibliography

- Allpress, C. J.; Grubel, K.; Szajna-Fuller, E.; Arif, A. M.; Berreau, L. M. Regioselective Aliphatic Carbon–Carbon Bond Cleavage by a Model System of Relevance to Iron-Containing Acireductone Dioxygenase. *J. Am. Chem. Soc.* 2013, 135 (2), 659–668. <https://doi.org/10.1021/ja3038189>.
- Bank, R. P. D. RCSB PDB - 1ZRR: Residual Dipolar Coupling Refinement of Acireductone Dioxygenase from *Klebsiella*. <https://www.rcsb.org/structure/1ZRR> (accessed 2025-12-12).
- Bank, R. P. D. RCSB PDB - 2HJI: Structural model for the Fe-containing isoform of acireductone dioxygenase. <https://www.rcsb.org/structure/2HJI> (accessed 2025-12-12).
- Bank, R. P. D. RCSB PDB - 4QGM: Acireductone dioxygenase from *Bacillus anthracis* with cadmium ion in active center. <https://www.rcsb.org/structure/4QGM> (accessed 2025-11-25).
- Bates, M. K.; Jia, Q.; Doan, H.; Liang, W.; Mukerjee, S. Charge-Transfer Effects in Ni–Fe and Ni–Fe–Co Mixed-Metal Oxides for the Alkaline Oxygen Evolution Reaction. *ACS Catal.* 2016, 6 (1), 155–161. <https://doi.org/10.1021/acscatal.5b01481>.
- Blade, G. A.; Parveen, R.; Jaimes, J. L.; Ilustre, W.; Saldaña, D.; Ivan, D. A.; Lynch, V. M.; Cundari, T. R.; Toledo, S. A Family of Structural and Functional Models for the Active Site of a Unique Dioxygenase: Acireductone Dioxygenase (ARD). *J Inorg Biochem* 2020, 212, 111253. <https://doi.org/10.1016/j.jinorgbio.2020.111253>.
- Chavez, F. A.; Banerjee, A.; Sljivic, B.; Chavez, F. A.; Banerjee, A.; Sljivic, B. Modeling the Metal Binding Site in Cupin Proteins. In *On Biomimetics*; IntechOpen, 2011. <https://doi.org/10.5772/18262>.

- Dai, Y.; Wensink, P. C.; Abeles, R. H. One Protein, Two Enzymes *. *Journal of Biological Chemistry* 1999, 274 (3), 1193–1195. <https://doi.org/10.1074/jbc.274.3.1193>.
- Deshpande, A. R.; Pochapsky, T. C.; Ringe, D. The Metal Drives the Chemistry: Dual Functions of Acireductone Dioxygenase. *Chem. Rev.* 2017, 117 (15), 10474–10501. <https://doi.org/10.1021/acs.chemrev.7b00117>.
- Deshpande, A. R.; Wagenpfeil, K.; Pochapsky, T. C.; Petsko, G. A.; Ringe, D. Metal-Dependent Function of a Mammalian Acireductone Dioxygenase. *Biochemistry* 2016, 55 (9), 1398–1407. <https://doi.org/10.1021/acs.biochem.5b01319>.
- Deshpande, A. R.; Wagenpfeil, K.; Pochapsky, T. C.; Petsko, G. A.; Ringe, D. Structure of Mouse Acireductone Dioxygenase with Ni²⁺ Ion and Pentanoic Acid in the Active Site: 5i8s, 2016. <https://doi.org/10.2210/pdb5i8s/pdb>.
- Fadul, A. Novel Iron (II) N₃O Biomimetic Model System of Relevance to CC Cleaving Dioxygenase Enzymes Including Human Acireductone Dioxygenase (ARD); American University, 2024.
- Hegg, E. L.; Que, L. The 2-His-1-Carboxylate Facial Triad--an Emerging Structural Motif in Mononuclear Non-Heme Iron(II) Enzymes. *Eur J Biochem* 1997, 250 (3), 625–629. <https://doi.org/10.1111/j.1432-1033.1997.t01-1-00625.x>.
- Ju, T.; Goldsmith, R. B.; Chai, S. C.; Maroney, M. J.; Pochapsky, S. S.; Pochapsky, T. C. One Protein, Two Enzymes Revisited: A Structural Entropy Switch Interconverts the Two Isoforms of Acireductone Dioxygenase. *Journal of Molecular Biology* 2006, 363 (4), 823–834. <https://doi.org/10.1016/j.jmb.2006.08.060>.

- Kal, S.; Que, L. Dioxygen Activation by Nonheme Iron Enzymes with the 2-His-1-Carboxylate Facial Triad That Generate High-Valent Oxoiron Oxidants. *J Biol Inorg Chem* 2017, 22 (2–3), 339–365. <https://doi.org/10.1007/s00775-016-1431-2>.
- Kirsch, K. E.; Little, M. E.; Cundari, T. R.; El-Shaer, E.; Barone, G.; Lynch, V. M.; Toledo, S. A. Direct O₂ Mediated Oxidation of a Ni(II)N₃O Structural Model Complex for the Active Site of Nickel Acireductone Dioxygenase (Ni-ARD): Characterization, Biomimetic Reactivity, and Enzymatic Implications. *Dalton Trans.* 2024, 53 (44), 17852–17863. <https://doi.org/10.1039/D4DT02538E>.
- Kuzmanic, Antonija, and Bojan Zagrovic. 2010. “Determination of Ensemble-Average Pairwise Root Mean-Square Deviation from Experimental B-Factors.” *Biophysical Journal* 98(5): 861–71. doi:10.1016/j.bpj.2009.11.011.
- Liu, X.; Garber, A.; Ryan, J.; Deshpande, A.; Ringe, D.; Pochapsky, T. C. A Model for the Solution Structure of Human Fe(II)-Bound Acireductone Dioxygenase and Interactions with the Regulatory Domain of Matrix Metalloproteinase I (MMP-I). *Biochemistry* 2020, 59 (44), 4238–4249. <https://doi.org/10.1021/acs.biochem.0c00724>.
- Miłaczewska, A.; Kot, E.; Amaya, J. A.; Makris, T. M.; Zając, M.; Korecki, J.; Chumakov, A.; Trzewik, B.; Kędracka-Krok, S.; Minor, W.; Chruszcz, M.; Borowski, T. On the Structure and Reaction Mechanism of Human Acireductone Dioxygenase. *Chemistry – A European Journal* 2018, 24 (20), 5225–5237. <https://doi.org/10.1002/chem.201704617>.
- Pochapsky, T. C.; Pochapsky, S. S.; Ju, T.; Hoefler, C.; Liang, J. A Refined Model for the Structure of Acireductone Dioxygenase from *Klebsiella* ATCC 8724 Incorporating

- Residual Dipolar Couplings. *J Biomol NMR* 2006, 34 (2), 117–127.
<https://doi.org/10.1007/s10858-005-5735-8>.
- Reilley, D. J.; Hennefarth, M. R.; Alexandrova, A. N. The Case for Enzymatic Competitive Metal Affinity Methods. *ACS Catal.* 2020, 10 (3), 2298–2307.
<https://doi.org/10.1021/acscatal.9b04831>.
- Sparta, M.; Valdez, C. E.; Alexandrova, A. N. Metal-Dependent Activity of Fe and Ni Acireductone Dioxygenases: How Two Electrons Reroute the Catalytic Pathway. *Journal of Molecular Biology* 2013, 425 (16), 3007–3018.
<https://doi.org/10.1016/j.jmb.2013.05.001>.
- Valdez, C. E.; Gallup, N. M.; Alexandrova, A. N. Co²⁺ Acireductone Dioxygenase: Fe²⁺ Mechanism, Ni²⁺ Mechanism, or Something Else? *Chemical Physics Letters* 2014, 604, 77–82. <https://doi.org/10.1016/j.cplett.2014.04.05>.
- Valdez, C. E.; Smith, Q. A.; Nechay, M. R.; Alexandrova, A. N. Mysteries of Metals in Metalloenzymes. *Acc Chem Res* 2014, 47 (10), 3110–3117.
<https://doi.org/10.1021/ar500227u>.
- Zhao, M.; Wang, H.-B.; Ji, L.-N.; Mao, Z.-W. Insights into Metalloenzyme Microenvironments: Biomimetic Metal Complexes with a Functional Second Coordination Sphere. *Chem. Soc. Rev.* 2013, 42 (21), 8360–8375.
<https://doi.org/10.1039/C3CS60162E>.